MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COUNTER, BINARY, DIGITAL, KEY CODE AMF

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic binary counter module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.0766 failures per million operating hours for class I or 0.3701 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be AMF.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/250-1 for class I and M28787/250-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{DD}) of 3.0 to 18.0 Vdc ±5%.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

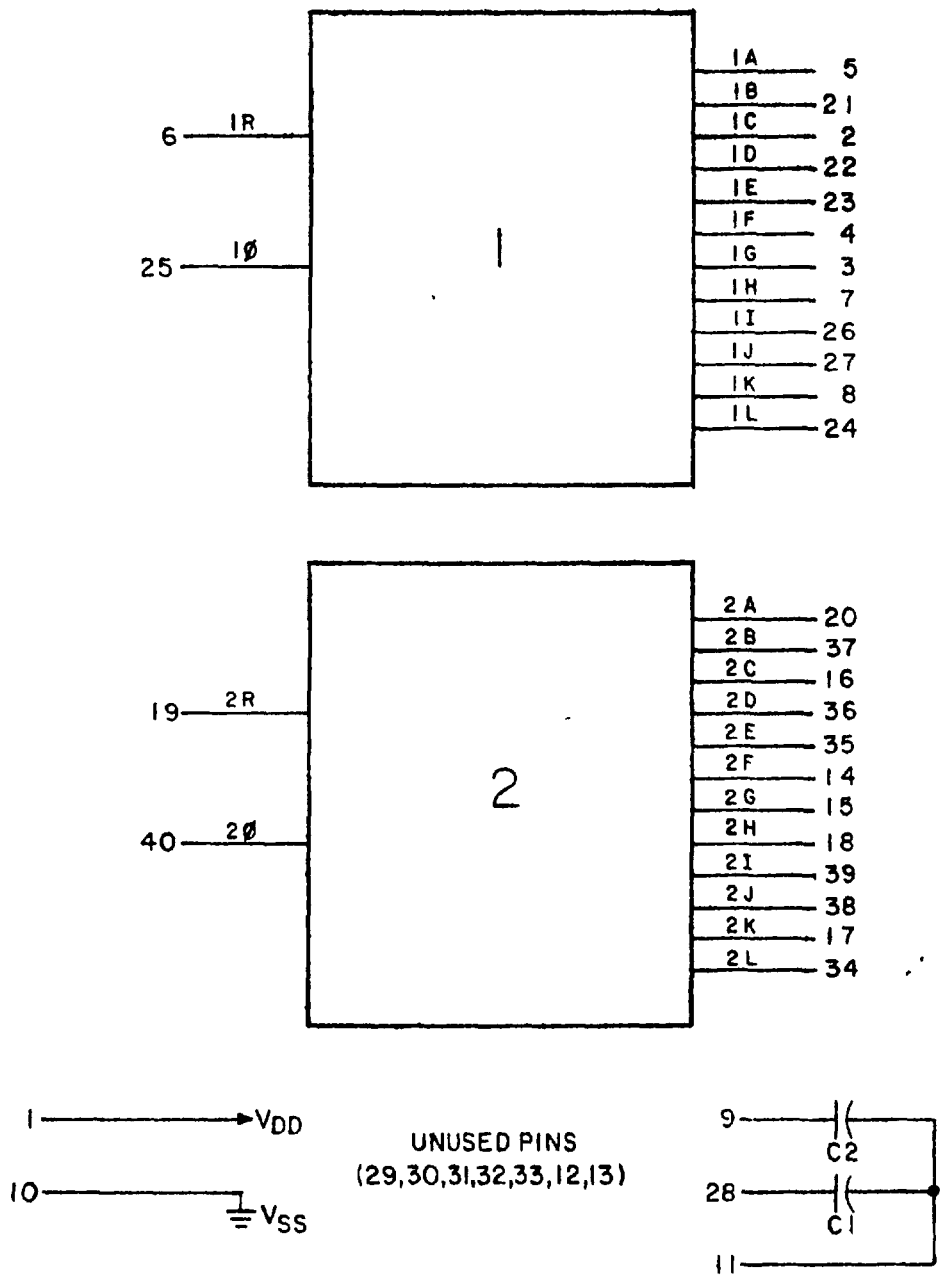


FIGURE 1. AMF module function diagram, binary counter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1,A2	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B4	Input clamp diode voltage, negative	IV	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C4	Input clamp diode voltage, positive	V	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D4096	Functional	VI	X	◄ See table VI ►										
E1-E24	Static output voltage, low level, (I_{OL} = 0), V_{DD} = 5 V	VII	X	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
F1-F24	Static output voltage, high level, (I_{OH} = 0), V_{DD} = 5 V	VIII	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
G1-G24	Static output voltage, low level, V_{DD} = 5 V	IX	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
H1-H24	Static output voltage, high level, V_{DD} = 5 V	X	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
J1-J24	Static output voltage, low level, V_{DD} = 15 V	XI	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
K1-K24	Static output voltage, high level, V_{DD} = 15 V	XII	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
L1-L4	Input current, low level	XIII	X	0	-200	---	---	0	-200	---	---	0	-200	nA
M1-M4	Input current, high level	XIV	X	0	200	---	---	0	200	---	---	0	200	nA
N1	Power supply current, low level	XV	X	0	40	0	40	0	580	0	40	0	1200	µA
N2	Power supply current, high level	XV	X	0	40	0	40	0	580	0	40	0	1200	µA
P1-P8	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	XVI	S	35	350	35	350	38	385	26	265	43	430	ns

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
P9-P16	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	XVI	Q	83	830	83	830	91	915	63	630	101	1015	ns
Q1-Q8	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	XVII	S	17	175	17	175	19	195	13	135	21	215	ns
Q9-Q16	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	XVII	Q	65	655	65	655	72	725	50	500	80	800	ns
R1-R8	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	XVIII	Q	15	150	15	150	16	165	11	115	18	185	ns
S1-S8	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	XIX	Q	7	75	7	75	8	85	5	55	9	90	ns
T1-T8	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	XX	Q	11	110	11	110	12	120	8	85	13	135	ns
U1-U8	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	XXI	Q	5	55	5	55	6	60	4	40	6	65	ns
V1,V5	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XXII	S	70	700	70	700	77	775	53	530	86	860	ns
V2-V4, V6-V8	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XXII	S	40	391	40	391	44	432	30	297	55	540	ns
V9,V13	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XXII	Q	90	900	90	900	99	995	68	685	110	1100	ns
V10-V12, V14-V16	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XXII	Q	60	591	60	591	66	653	44	449	73	724	ns
W1,W5	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XXIII	S	70	700	70	700	77	775	53	530	86	860	ns
W2-W4, W6-W8	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XXIII	S	40	391	40	391	44	432	30	297	55	540	ns

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
W9,W13	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XXIII	Q	90	900	90	900	99	995	68	685	110	1100	ns
W10-W12, W14-W16	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XXIII	Q	60	591	60	591	66	553	44	449	73	724	ns
X1,X5	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXIV	Q	28	280	28	280	31	310	21	210	34	345	ns
X2-X4, X6-X8	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXIV	Q	14	138	14	138	16	153	11	105	17	169	ns
Y1,Y5	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXV	Q	28	280	28	280	31	310	21	210	34	345	ns
Y2-Y4, Y6-Y8	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXV	Q	14	138	14	138	16	153	11	105	17	169	ns
Z1,Z5	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXVI	Q	21	210	21	210	23	230	16	160	25	255	ns
Z2-Z4, Z6-Z8	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXVI	Q	11	108	11	108	12	120	8	82	14	133	ns
AA1,AA5	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXVII	Q	21	210	21	210	23	230	16	160	25	255	ns
AA2-AA4, AA6-AA8	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXVII	Q	11	108	11	108	12	120	8	82	14	133	ns
BB1-BB24	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XXVIII	S	108	1080	108	1080	119	1195	82	820	132	1325	ns
BB25-BB48	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XXVIII	Q	128	1280	128	1280	141	1415	97	975	157	1570	ns

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
CC1-CC24	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXIX	Q	40	400	40	400	44	445	30	305	49	490	ns
DD1-DD24	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXX	Q	30	300	30	300	33	330	23	230	37	370	ns
EE	Minimum reset pulse width check	XXXI	S	← See table XXXI →										
FF1,FF2	Maximum clock frequency check	XXXII	S	1.95	2.15	1.95	2.15	1.95	2.15	1.95	2.15	1.95	2.15	ms
GG	Power supply filter capacitance	XXXIII	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
HH1,HH2	Capacitance C1 and C2	XXXIV	X	176	264	176	264	176	264	176	264	176	264	pF
JJ1-JJ4	Input capacitance	XXXV	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through JJ shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through JJ in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through K, N through DD, and HH shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition GG, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B through K, N through DD and HH shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through JJ shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 100°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation	18	18	mW
Input voltage			
Low level, V_{DD} = 5 V	0.0 min, 1.5 max	0.0 min, 1.5 max	V
V_{DD} = 15 V	0 0 min, 4.0 max	0.0 min, 4.0 max	V
High level, V_{DD} = 5 V	3.5 min, V_{DD} max	3.5 min, V_{DD} max	V
V_{DD} = 15 V	11 min, V_{DD} max	11 min, V_{DD} max	V
Input current, (V_{DD} = 18 V)			
Low level (V_{IL} = 0 V)	0 min, -200 max	0 min, -200 max	nA
High level (V_{IH} = 18 V)	0 min, 200 max	0 min, 200 max	nA
Output drive current (V_{DD} = 5 V)			
Low level (V_{OL} = 0.4 V)			
I_{OL} at -55°C	---	0.64 min	mA
0° and 25°C	0.51 min	0.51 min	mA
60°C	0.44 min	0.44 min	mA
100°C	---	0.36 min	mA
High level (V_{OH} = 4.6 V)			
I_{OH} at -55°C	---	-0.64 min	mA
0° and 25°C	-0.51 min	-0.51 min	mA
60°C	-0.44 min	-0.44 min	mA
100°C	---	-0.36 min	mA
Clock input			
Maximum frequency, V_{DD} = 5 V	2.0	2.0	MHz
V_{DD} = 10 V	5.0	5.0	MHz
V_{DD} = 15 V	6.5	6.5	MHz
Minimum clock pulse width, V_{DD} = 5 V	250	250	ns
V_{DD} = 10 V	100	100	ns
V_{DD} = 15 V	75	75	ns
Minimum reset pulse width, V_{DD} = 5 V	640	640	ns
V_{DD} = 10 V	240	240	ns
V_{DD} = 15 V	180	180	ns

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

6.5 Module function. The modules contains two independent 12-bit binary counters and two 220 pF capacitors, with one side connected to pin 11 for external use.

6.5.1 Input to output relationship. Each gate shall have the input to output relationship of table II.

TABLE II. Input to output relationship. 1/-5/

Input		Output
Clock	Reset	Output state
⎍	L	No change
⎎	L	Advance to next state
X	H	All outputs are low

1/ X = Don't care.

2/ L = Low level.

3/ H = High level.

4/ ⎍ = Positive going edge of clock.

5/ ⎎ = Negative going edge of clock.

6.5.2 Paralleling of outputs. Paralleling of outputs is not permissible.

6.5.3 Dynamic power dissipation. The dynamic power dissipation can be approximated by the formula:

$$P_{TOTAL} = (C\ V_{DD}^2 f)\ N + \text{Quiescent power dissipation}$$

Where: C = Load capacitance.

V_{DD} = Supply voltage.

f = Operating frequency.

N = Number of outputs.

6.6 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed:

a. Device pins should be in contact with conductive material except when being tested or in actual operation, to avoid build-up of static charge.

b. Soldering iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.

c. Signals should not be applied to the inputs while the device power supply (V_{DD}) is off.

d. All input signals and conditioning voltages must be removed before V_{DD} is disconnected.

e. Input voltages greater than the supply (V_{DD} - V_{SS}) should not be applied as this will forward bias the protection input diodes.

f. All unused input leads must be connected to either V_{SS} (GND) or V_{DD} (device supply), whichever is appropriate for the logic circuit involved.

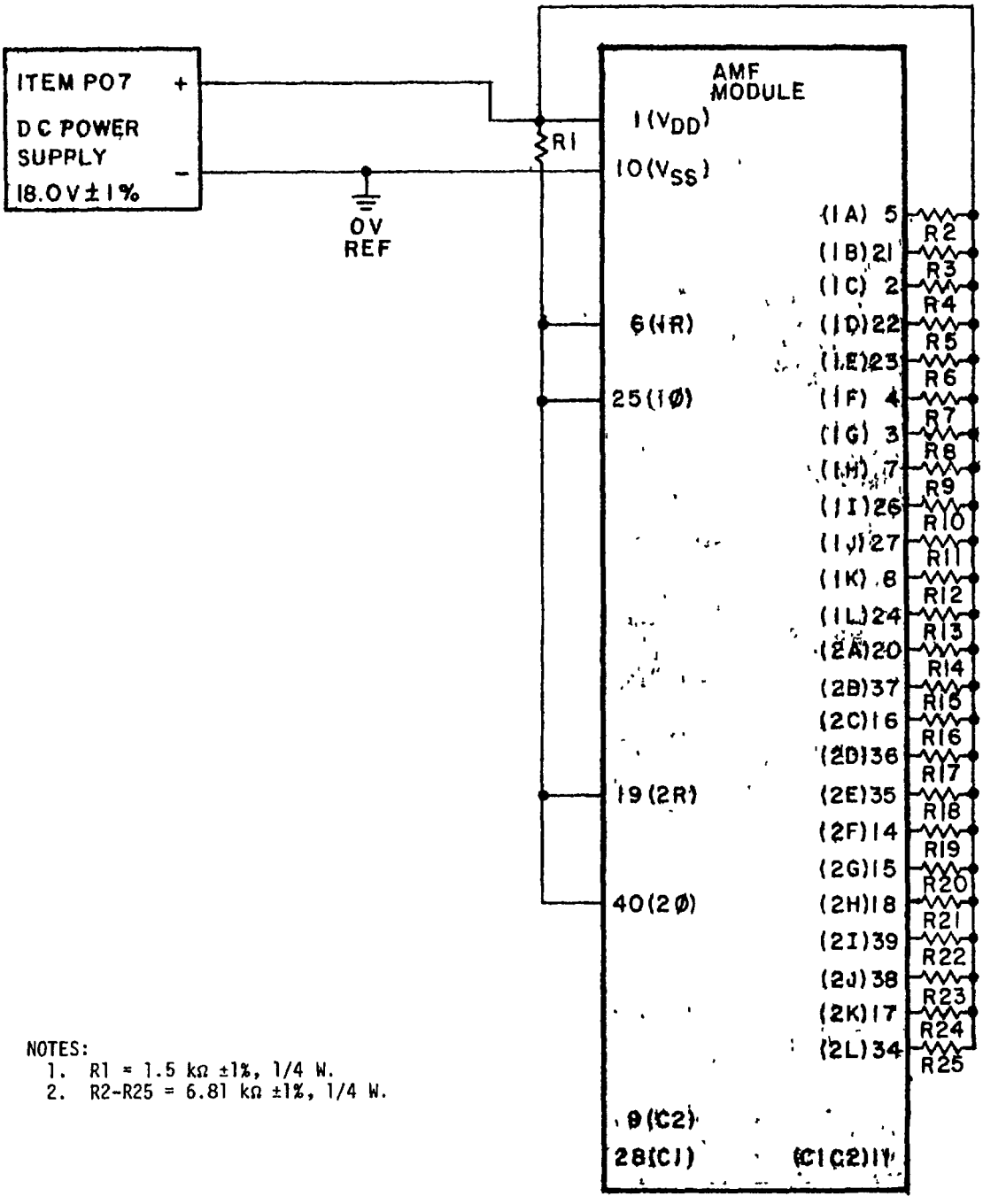
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N258)

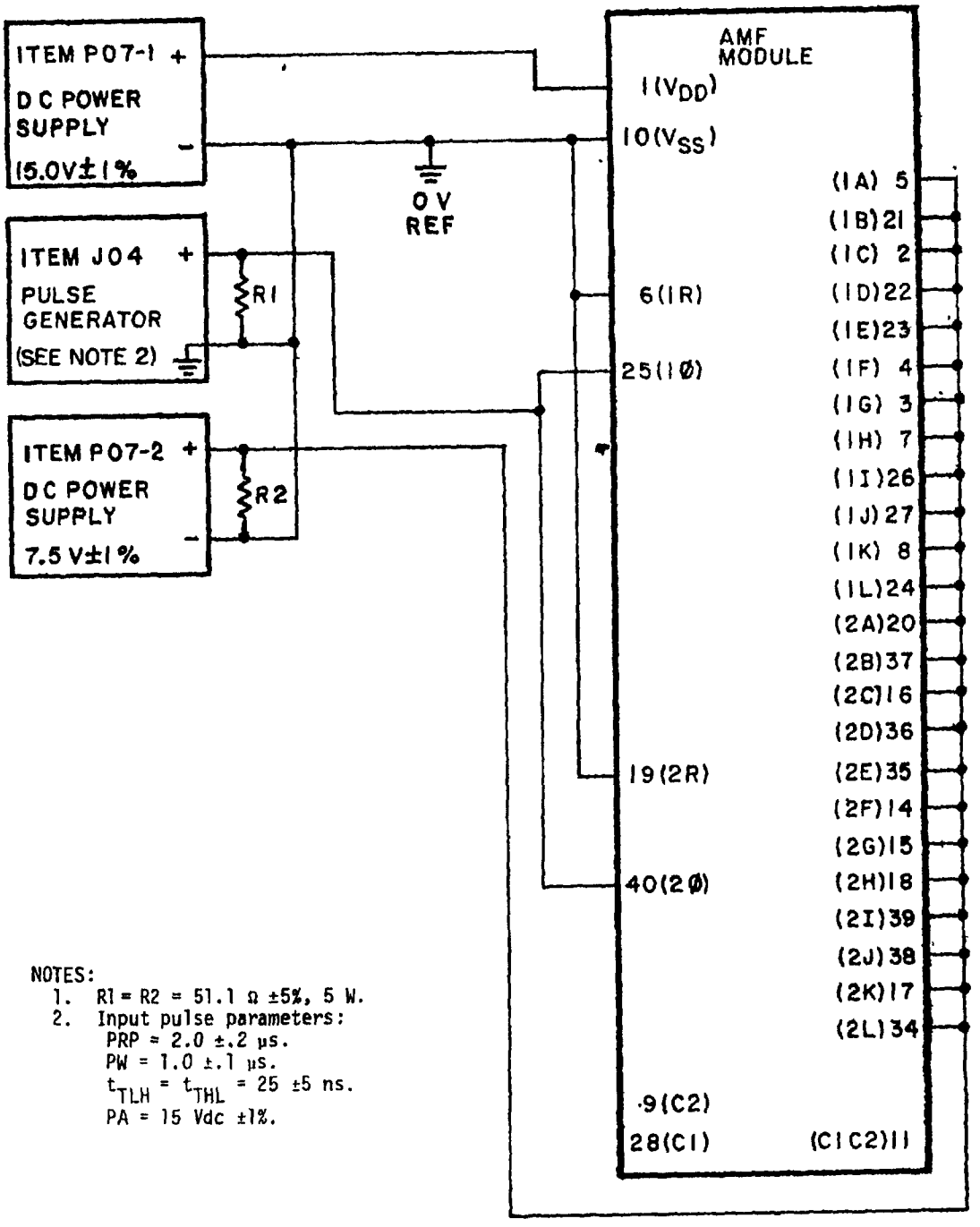
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 1.5 kΩ ±1%, 1/4 W.
2. R2-R25 = 6.81 kΩ ±1%, 1/4 W.

FIGURE 2 Operating temperature and life test circuit

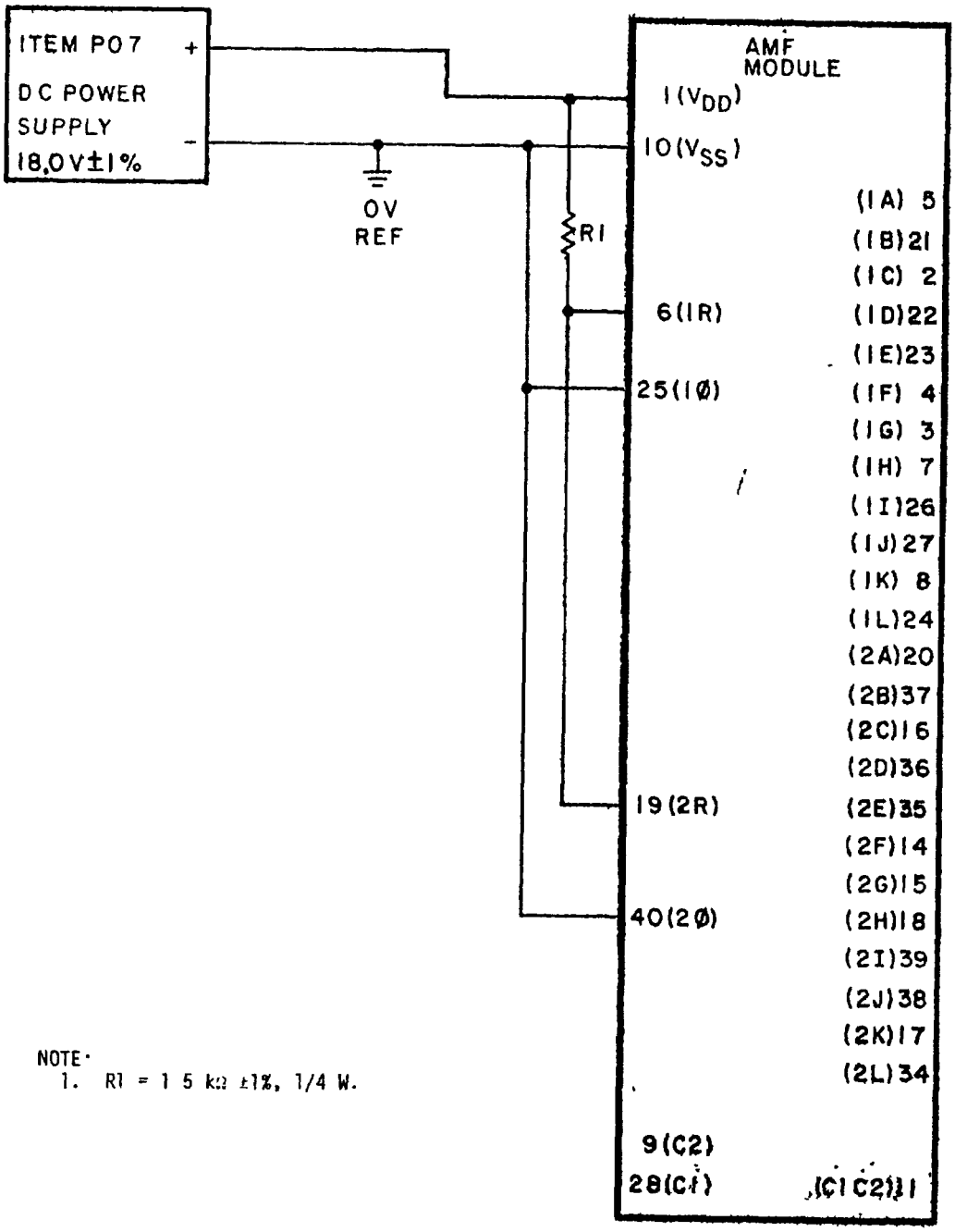
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51.1 Ω ±5%, 5 W.
2. Input pulse parameters:
   PRP = 2.0 ±.2 µs.
   PW = 1.0 ±.1 µs.
   $t_{TLH} = t_{THL}$ = 25 ±5 ns.
   PA = 15 Vdc ±1%.

FIGURE 3. Component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTE:
1. R1 = 1 5 kΩ ±1%, 1/4 W.

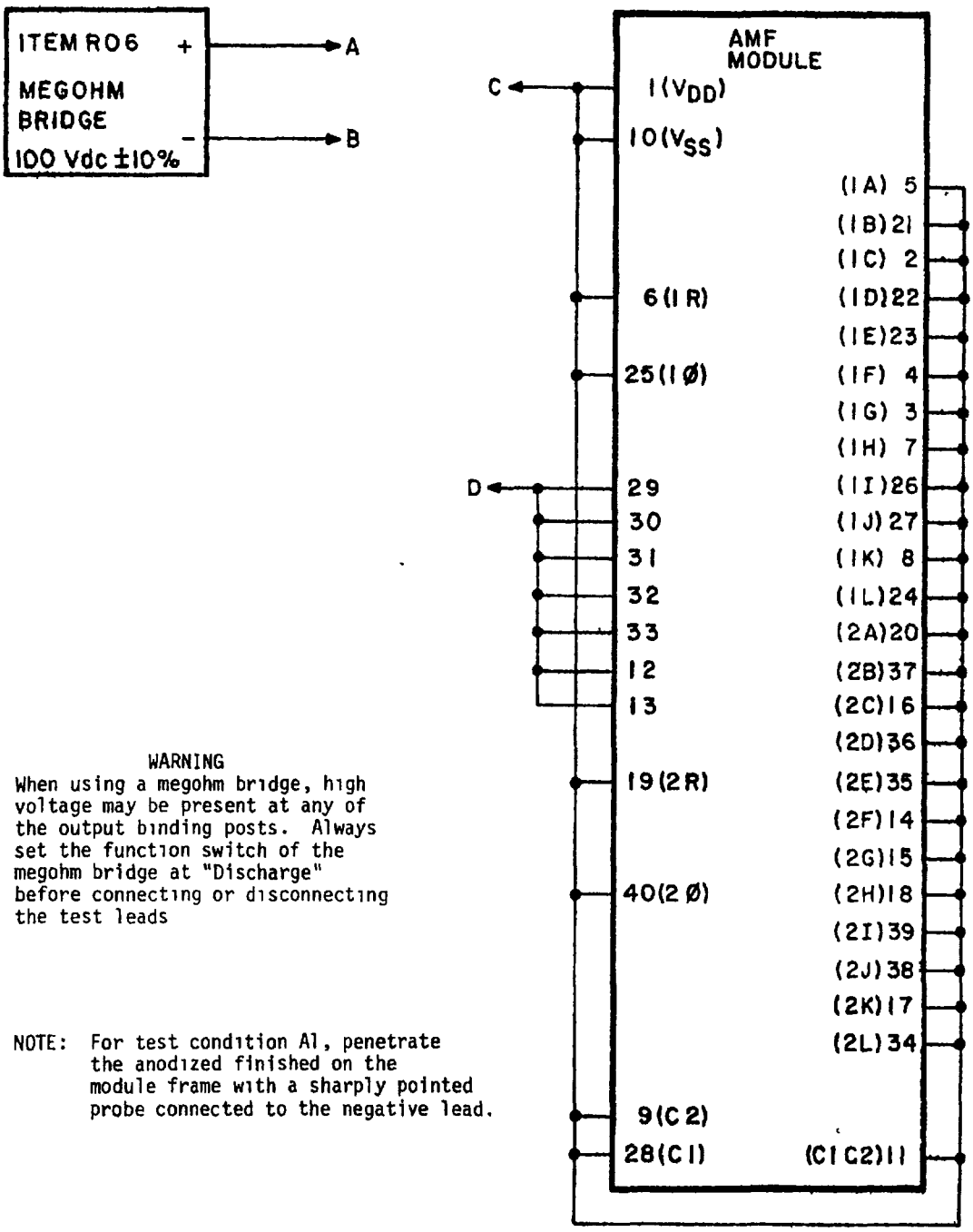
FIGURE 4 Vibration test circuit

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C, D	Frame	10	∞	MΩ	5
A2	Isolation, unused pins	C	D	10	∞	MΩ	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



WARNING
When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge at "Discharge" before connecting or disconnecting the test leads

NOTE: For test condition A1, penetrate the anodized finished on the module frame with a sharply pointed probe connected to the negative lead.

FIGURE 5 Isolation test circuit for table III

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

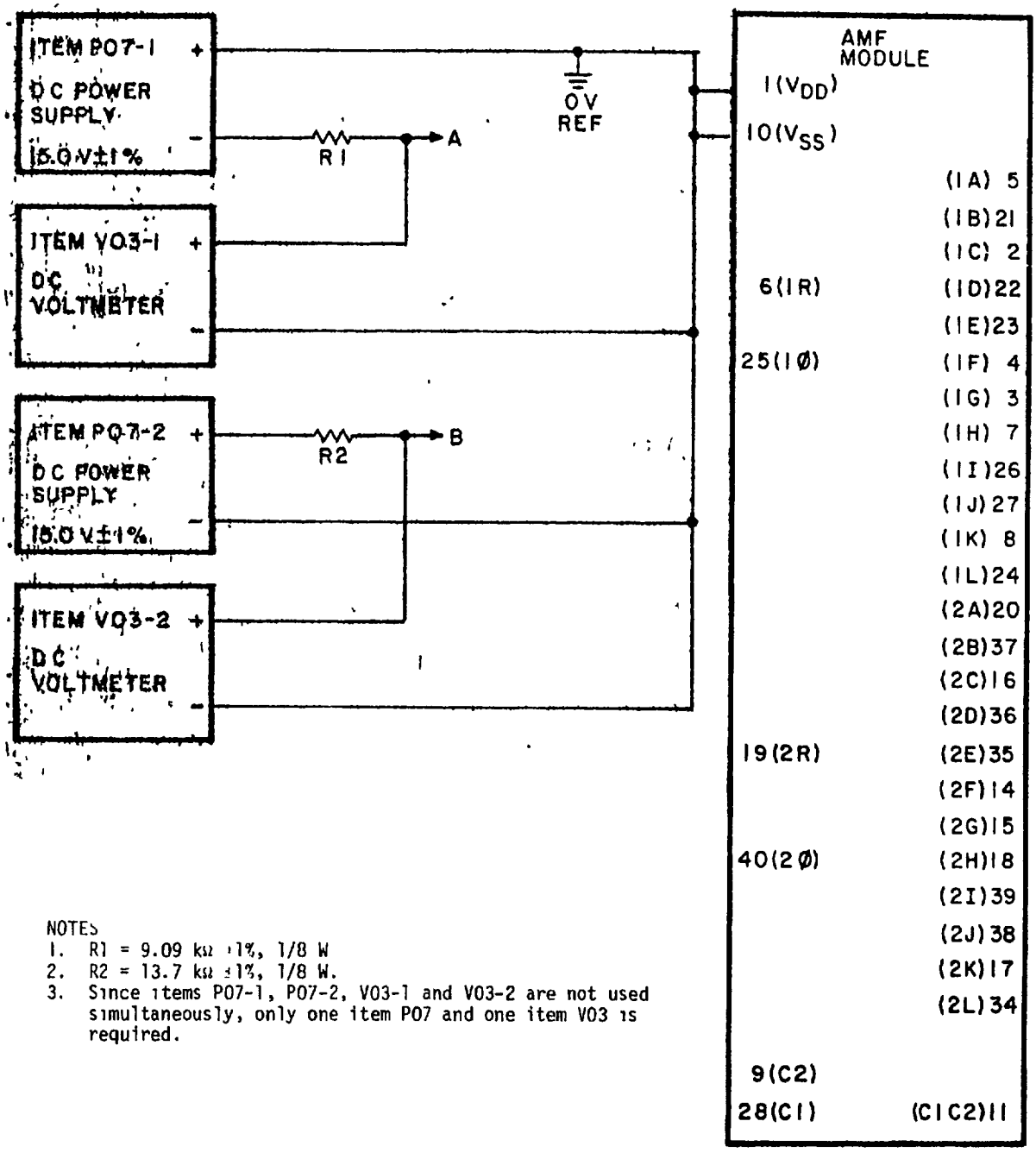
TABLE IV. Input clamp diode voltage (negative) test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B1	1R	Input clamp diode voltage, negative	6	Open	-0.4	-6.0	V	6
B2	1Ø	↓	25	↓	↓	↓	↓	↓
B3	2R		19					
B4	2Ø		40					

TABLE V. Input clamp diode voltage (positive) test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	1R	Input clamp diode voltage, positive	Open	6	0.4	1.5	V	6
C2	1Ø	↓	↓	25	↓	↓	↓	↓
C3	2R			19				
C4	2Ø			40				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES
1. R1 = 9.09 kΩ ±1%, 1/8 W
2. R2 = 13.7 kΩ ±1%, 1/8 W.
3. Since items PO7-1, PO7-2, VO3-1 and VO3-2 are not used simultaneously, only one item PO7 and one item VO3 is required.

FIGURE 6 Input clamp diode voltage (negative and positive) test circuit for tables IV and V

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE VI. Functional test.

Test condition	Parameter	Output states 1/ - 4/																								No. of pulses	Test figure
		1A	1B	1C	1D	1E	1F	1G	1H	1I	1J	1K	1L	2A	2B	2C	2D	2E	2F	2G	2H	2I	2J	2K	2L		
D1	Functional	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	0	7,8
D2		H	L	L	L	L	L	L	L	L	L	L	L	H	L	L	L	L	L	L	L	L	L	L	L	1	
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
D4096	↓	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	↓	↓

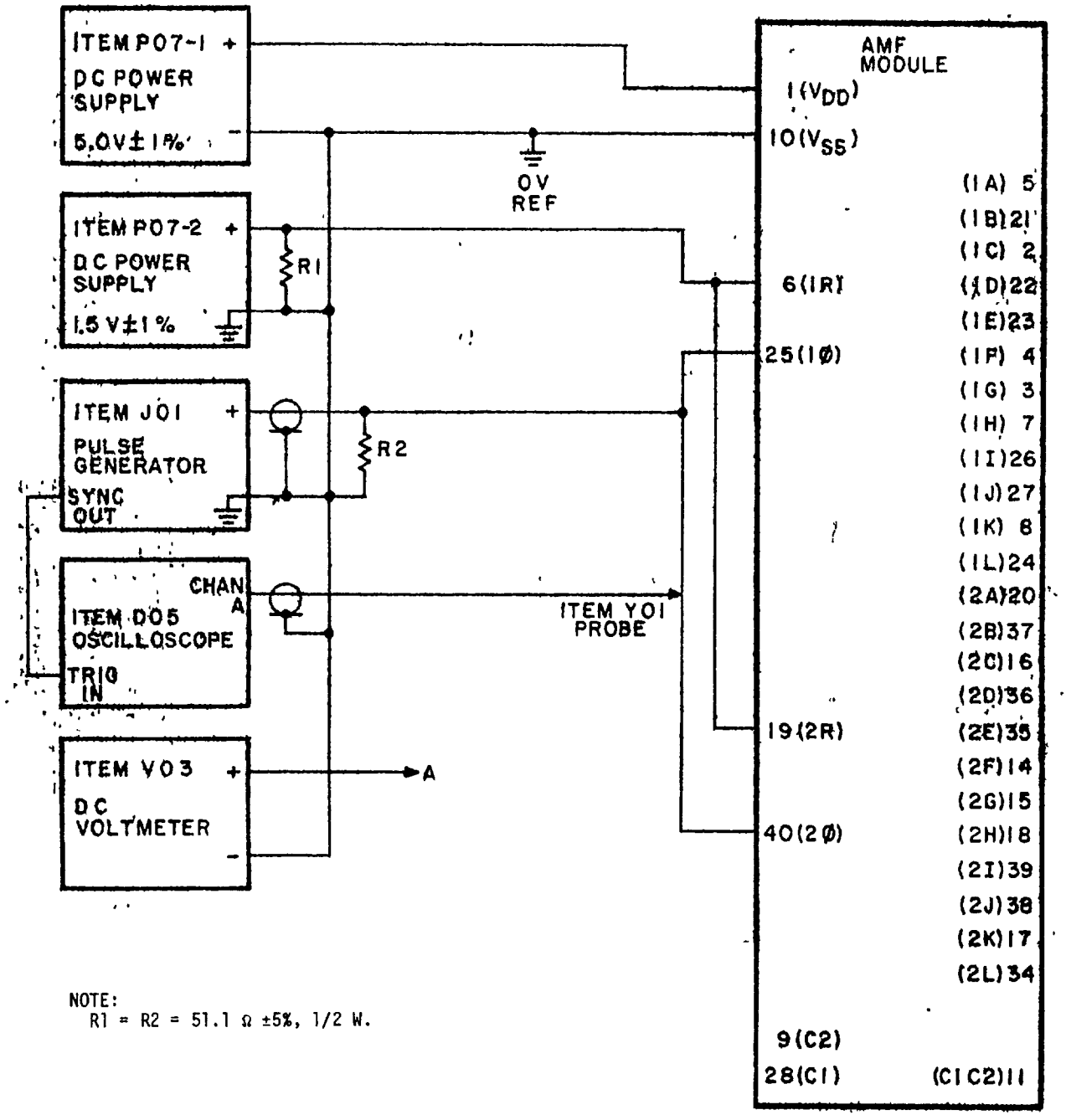
1/ Output logic states: L = low level = 0.0 to 0.4 V.
H = high level = 4.6 to 5.0 V.

2/ Counters 1 and 2 with outputs 1A to 1L and 2A to 2L each shall have a binary output code with test conditions D1 to D4096 corresponding to that binary code. Test condition D1 is the initial state of the code and test condition D4096 is the final state of the code.

3/ For test conditions D1-D4096, verification of the output state is obtained by connecting voltmeter VO3, point A, to each output pin in succession after input conditions have been accomplished.

4/ 1A-1L and 2A-2L output logic states are measured at output pins 5, 21, 2, 22, 23, 4, 3, 7, 26, 27, 8, 24, and 20, 37, 16, 36, 35, 14, 15, 18, 33, 38, 17, 34; respectively.

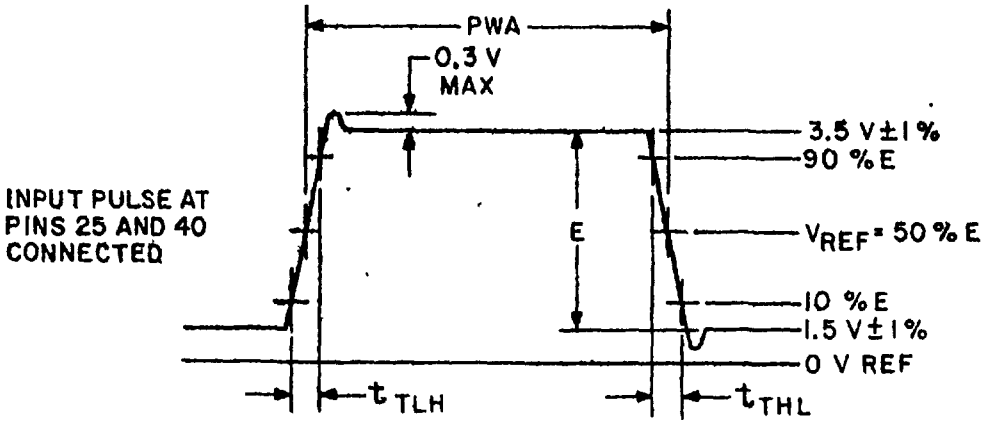
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTE:
R1 = R2 = 51.1 Ω ±5%, 1/2 W.

FIGURE 7. Functional test circuit for table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PWA = 400 ±20 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Immediately before starting test condition D1, momentarily apply pins 6 and 19 to V_{DD}, then with item J01 in the single pulse mode apply one pulse for test conditions D2-D4096 and then take the measurement.
3. Test conditions D1-D4096 should be done in sequence with no power interruptions to V_{DD}.
4. All voltages are referenced to ground

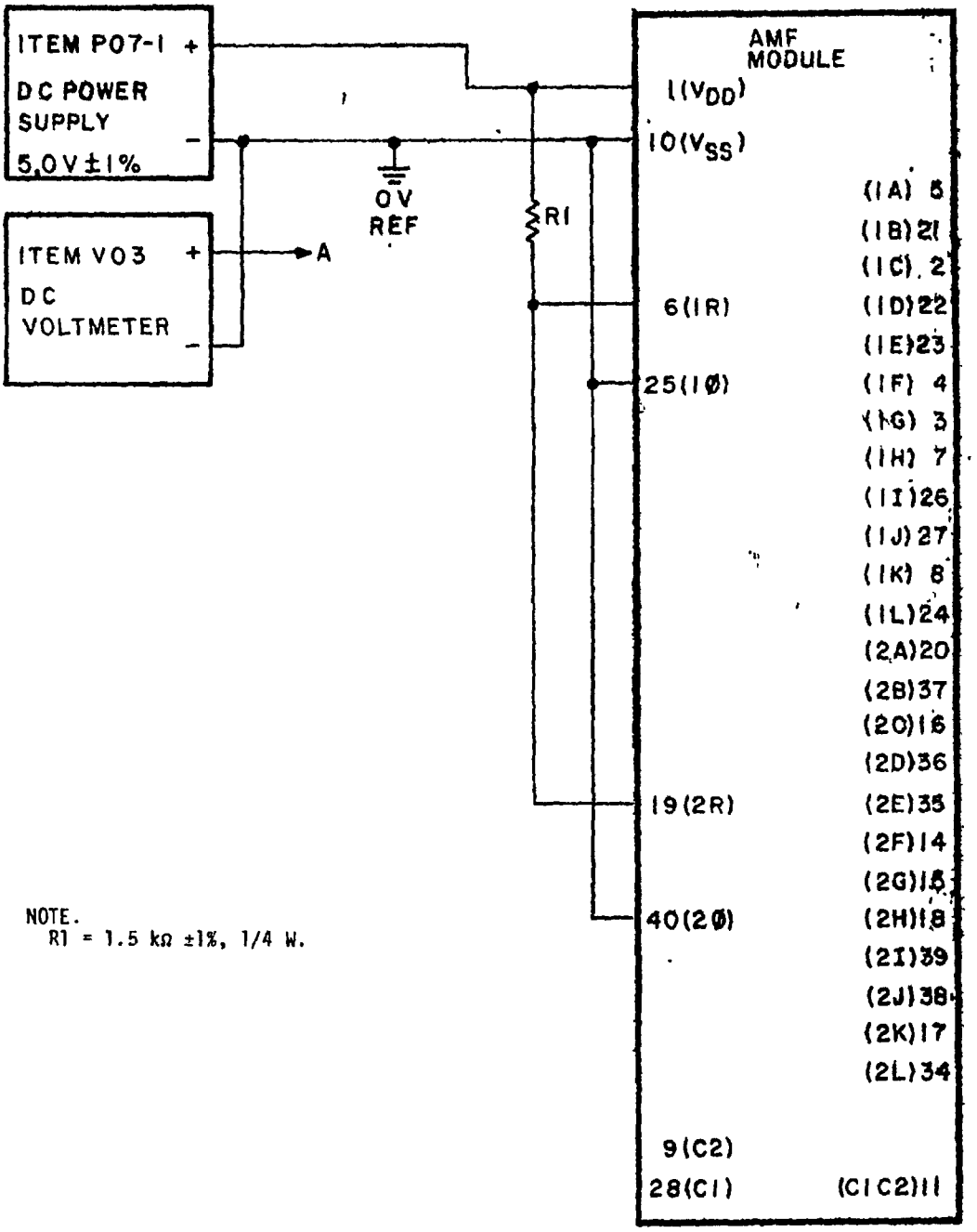
FIGURE 8. Functional test circuit waveform for table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE VII. Static output voltage (low-level) test.

Test condition		Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
E1	1A	Static output voltage, low level, (I_{OL} = 0), V_{DD} = 5 V	5	0.00	0.05	V	9
E2	1B		21				
E3	1C		2				
E4	1D		22				
E5	1E		23				
E6	1F		4				
E7	1G		3				
E8	1H		7				
E9	1I		26				
E10	1J		27				
E11	1K		8				
E12	1L		24				
E13	2A		20				
E14	2B		37				
E15	2C		16				
E16	2D		36				
E17	2E		35				
E18	2F		14				
E19	2G		15				
E20	2H		18				
E21	2I		39				
E22	2J		38				
E23	2K		17				
E24	2L		34				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTE.
R1 = 1.5 kΩ ±1%, 1/4 W.

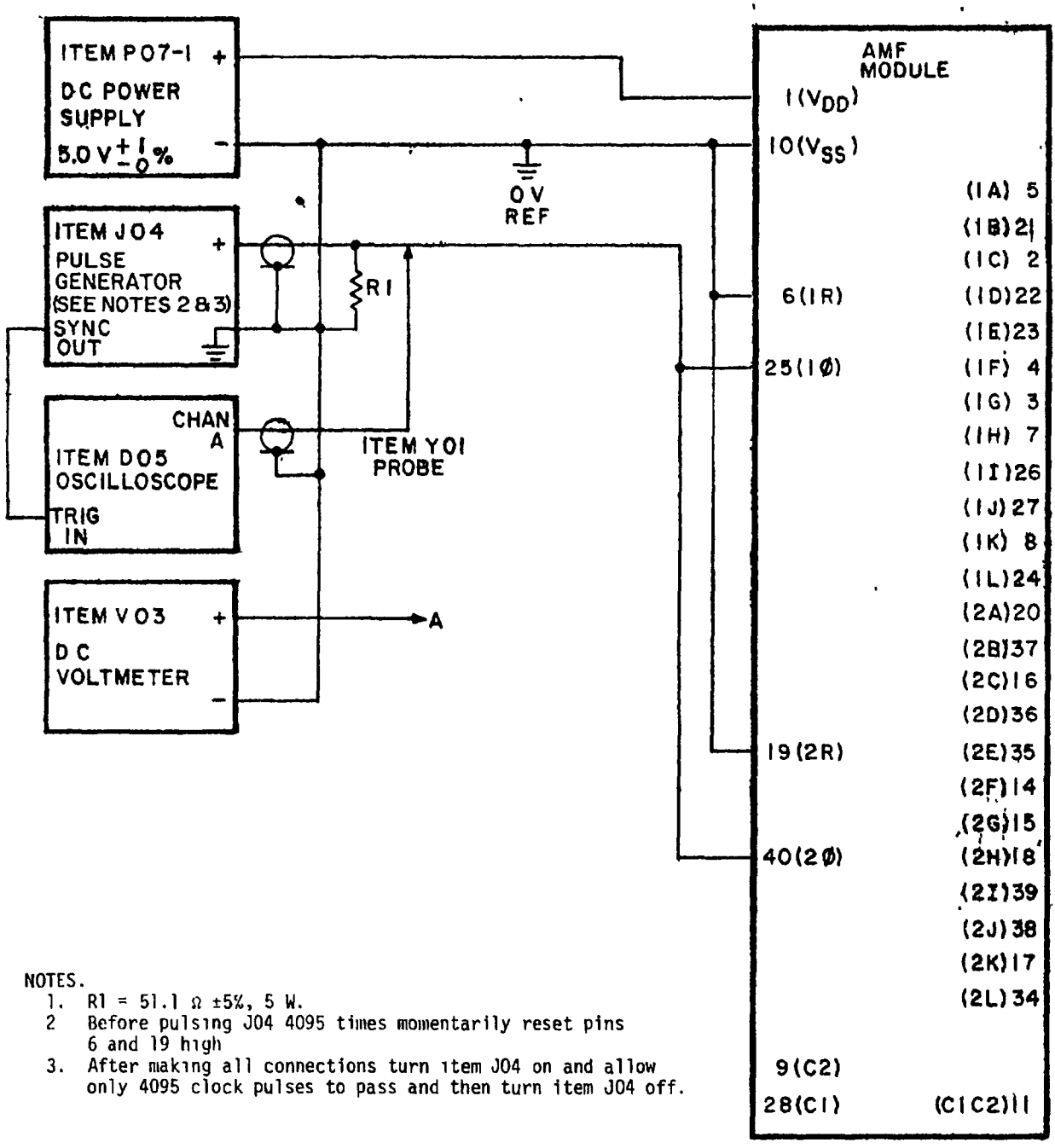
FIGURE 9. Static output voltage (low-level) test circuit for table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE VIII. Static output voltage (high-level) test.

Test condition		Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
F1	1A	Static output voltage, high level, (I_{OH} = 0), V_{DD} = 5 V	5	4.95	5.05	V	10,11
F2	1B		21				
F3	1C		2				
F4	1D		22				
F5	1E		23				
F6	1F		4				
F7	1G		3				
F8	1H		7				
F9	1I		26				
F10	1J		27				
F11	1K		8				
F12	1L		24				
F13	2A		20				
F14	2B		37				
F15	2C		16				
F16	2D		36				
F17	2E		35				
F18	2F		14				
F19	2G		15				
F20	2H		18				
F21	2I		39				
F22	2J		38				
F23	2K		17				
F24	2L		34				

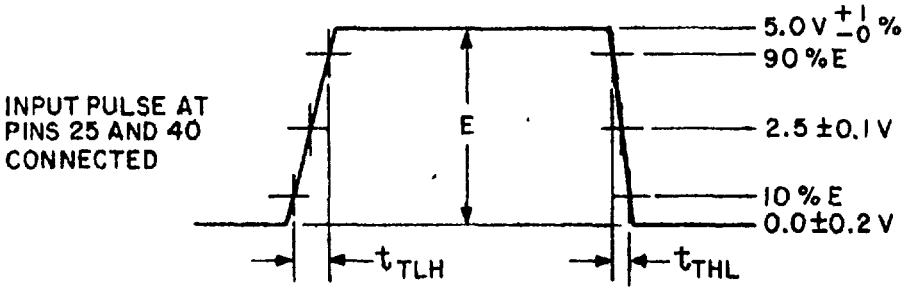
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES.
1. R1 = 51.1 Ω ±5%, 5 W.
2. Before pulsing J04 4095 times momentarily reset pins 6 and 19 high
3. After making all connections turn item J04 on and allow only 4095 clock pulses to pass and then turn item J04 off.

FIGURE 10. Static output voltage (high-level) test circuit for table VIII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters
   PWA = 1000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. All voltages are referenced to ground.

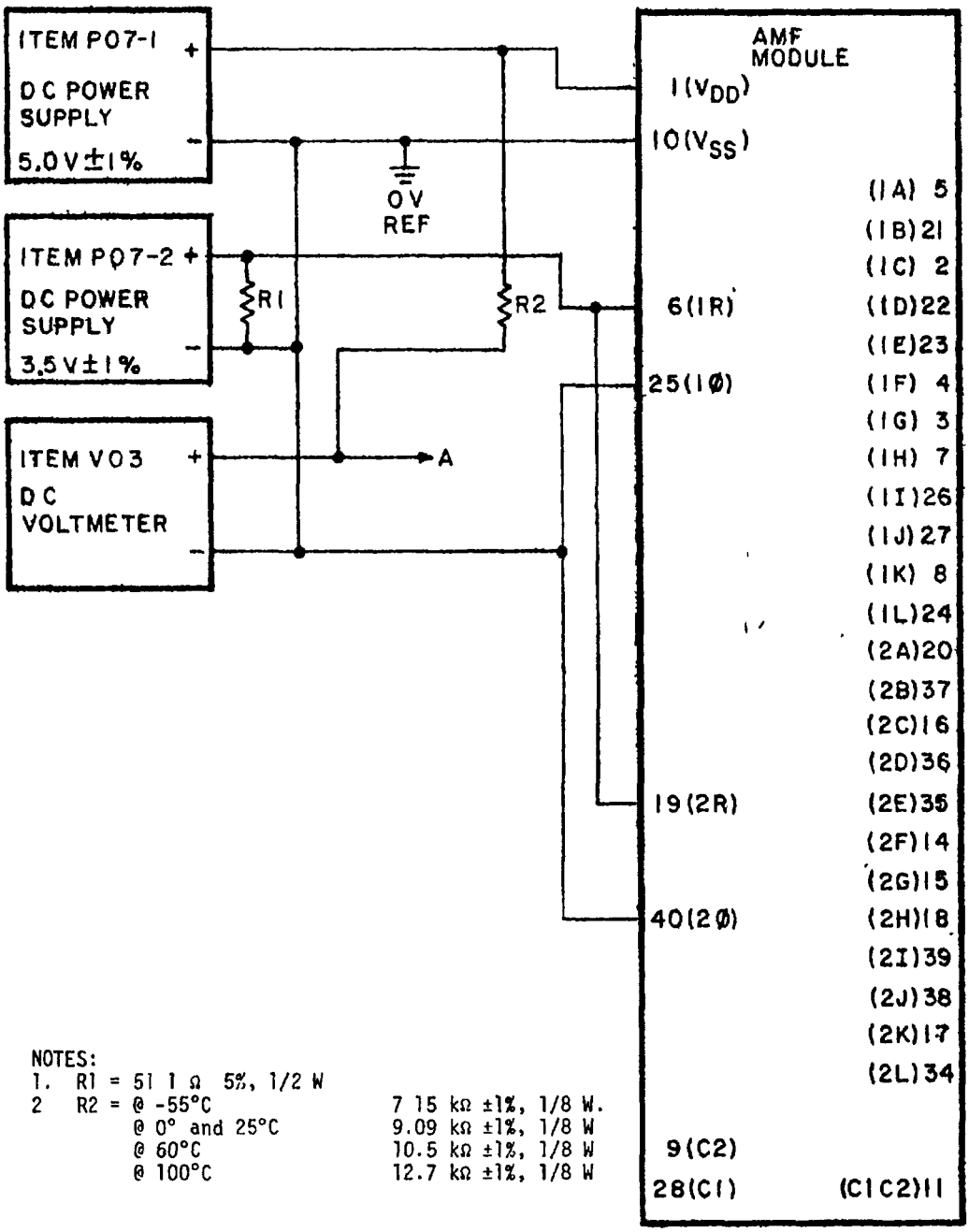
FIGURE 11. Static output voltage (high-level) test circuit waveform for table VIII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE IX. Static output voltage (low-level, V_{DD} = 5 V) test.

Test condition		Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
G1	1A	Static output voltage, low level, V_{DD} = 5 V	5	0.0	0.4	V	12
G2	1B		21				
G3	1C		2				
G4	1D		22				
G5	1E		23				
G6	1F		4				
G7	1G		3				
G8	1H		7				
G9	1I		26				
G10	1J		27				
G11	1K		8				
G12	1L		24				
G13	2A		20				
G14	2B		37				
G15	2C		16				
G16	2D		36				
G17	2E		35				
G18	2F		14				
G19	2G		15				
G20	2H		18				
G21	2I		39				
G22	2J		38				
G23	2K		17				
G24	2L		34				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51 1 Ω 5%, 1/2 W
2 R2 = @ -55°C 7 15 kΩ ±1%, 1/8 W.
@ 0° and 25°C 9.09 kΩ ±1%, 1/8 W
@ 60°C 10.5 kΩ ±1%, 1/8 W
@ 100°C 12.7 kΩ ±1%, 1/8 W

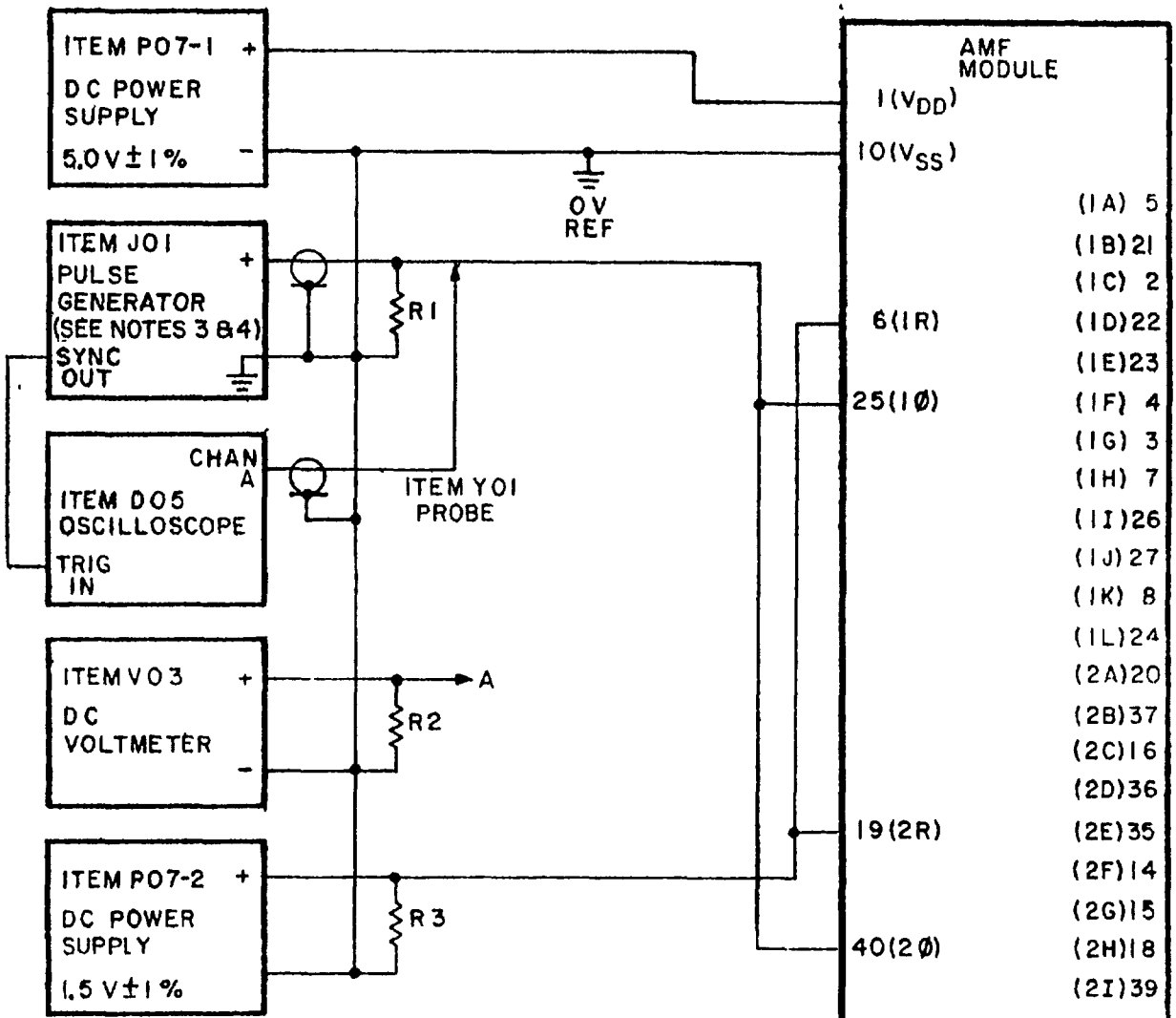
FIGURE 12. Static output voltage (low-level, V_{DD} = 5.0 V) test circuit for table IX

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE X. Static output voltage (high-level, V_{DD} = 5 V) test.

Test condition		Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
H1	1A	Static output voltage, high level, V_{DD} = 5 V	5	4.6	5.0	V	13,14
H2	1B		21				
H3	1C		2				
H4	1D		22				
H5	1E		23				
H6	1F		4				
H7	1G		3				
H8	1H		7				
H9	1I		26				
H10	1J		27				
H11	1K		8				
H12	1L		24				
H13	2A		20				
H14	2B		37				
H15	2C		16				
H16	2D		36				
H17	2E		35				
H18	2F		14				
H19	2G		15				
H20	2H		18				
H21	2I		39				
H22	2J		38				
H23	2K		17				
H24	2L		34				

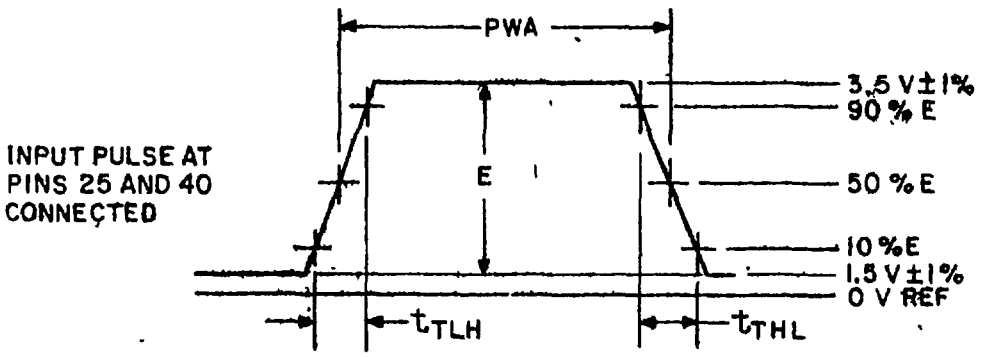
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1, R3 = 51.1 Ω ±5%, 1/2 W.
2. R2 = @ -55°C 18.2 kΩ ±1%, 1/8 W
   @ 0° and 25°C 23.2 kΩ ±1%, 1/8 W
   @ 60°C 27.4 kΩ ±1%, 1/8 W.
   @ 100°C 32 4 kΩ ±1%, 1/8 W.
3. Before pulsing J01 4095 times, momentarily reset pins 6 and 19 high.
4. After making all connections, turn item J01 on and allow only 4095 clock pulses to pass and then turn item J01 off

FIGURE 13 Static output voltage (high-level, V_{DD} = 5 V) test circuit for table X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PWA = 1000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. All voltages are referenced to ground.

FIGURE 14. Static output voltage (high-level, V_{DD} = 5 V) test circuit waveform for table X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XI. Static output voltage (low-level, V_{DD} = 15 V) test.

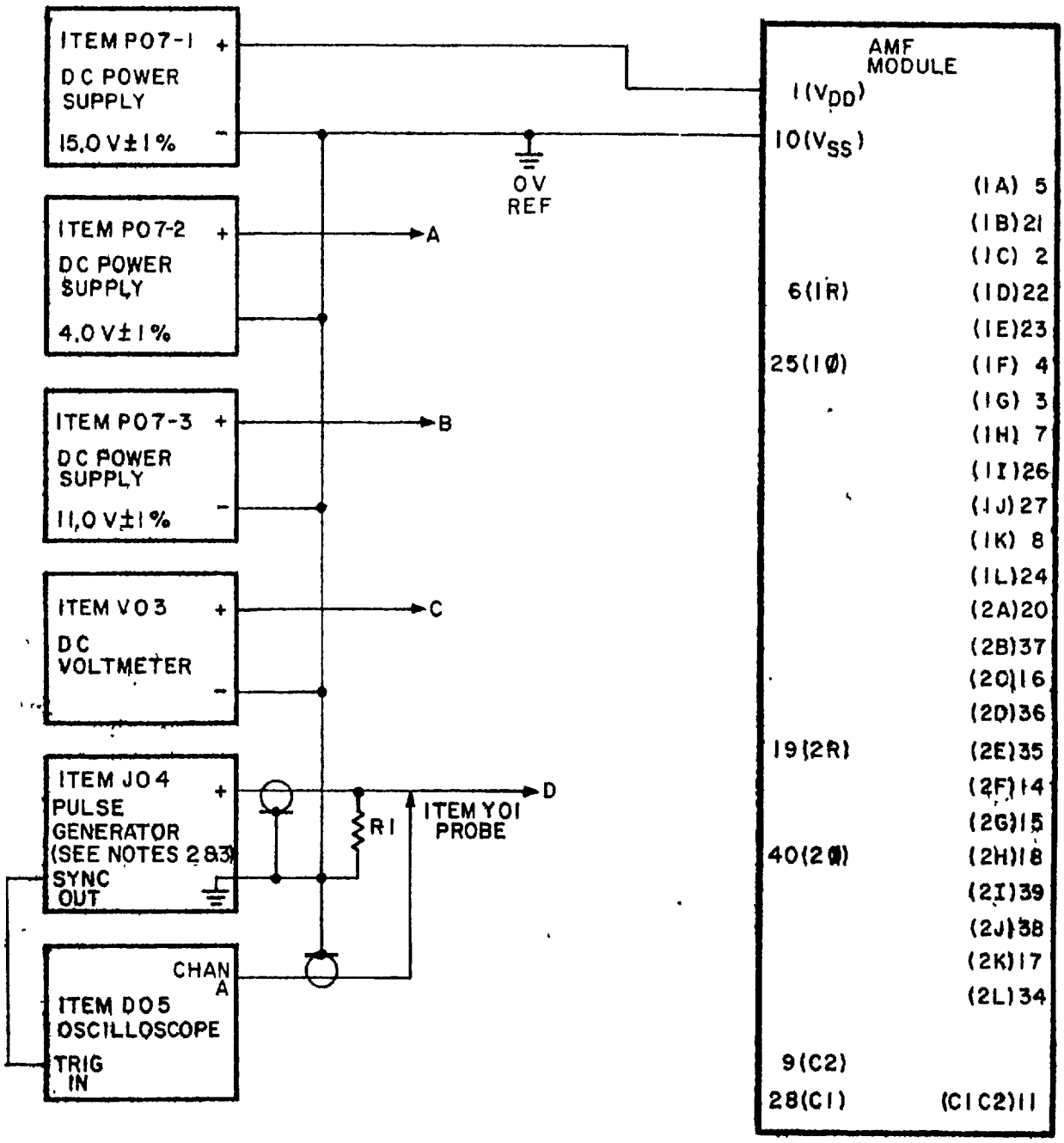
Test condition		Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
J1	1A	Static output voltage, low level, V_{DD} = 15 V	25,40	6,19	5	Open	0.0	1.5	V	15,16
J2	1B				21					
J3	1C				2					
J4	1D				22					
J5	1E				23					
J6	1F				4					
J7	1G				3					
J8	1H				7					
J9	1I				26					
J10	1J				27					
J11	1K				8					
J12	1L				24					
J13	2A				20					
J14	2B				37					
J15	2C				16					
J16	2D				36					
J17	2E				35					
J18	2F				14					
J19	2G				15					
J20	2H				18					
J21	2I				39					
J22	2J				38					
J23	2K				17					
J24	2L				34					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XII. Static output voltage (high-level, V_{DD} = 15 V) test.

Test condition		Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
K1	1A	Static output voltage, high level, V_{DD} = 15 V	6,19	Open	5	25,40	13.5	15.0	V	15,16
K2	1B				21					
K3	1C				2					
K4	1D				22					
K5	1E				23					
K6	1F				4					
K7	1G				3					
K8	1H				7					
K9	1I				26					
K10	1J				27					
K11	1K				8					
K12	1L				24					
K13	2A				20					
K14	2B				37					
K15	2C				16					
K16	2D				36					
K17	2E				35					
K18	2F				14					
K19	2G				15					
K20	2H				18					
K21	2I				39					
K22	2J				38					
K23	2K				17					
K24	2L				34					

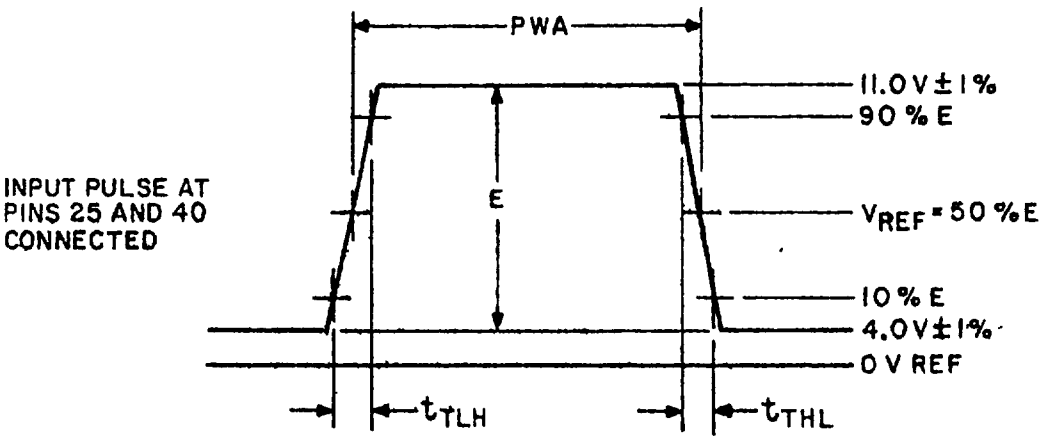
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51.1Ω ±5%, 5 W
2. For test conditions K1-K24 momentarily reset pins 6 and 19 high before connecting point A and pulsing J04 4095 times.
3. For test conditions K1-K24 allow only 4095 clock pulses to pass and then turn item J04 off before making measurements.
4. All voltages are referenced to ground.

FIGURE 15. Static output voltage (low-level and high-level, V_{DD} = 15 V) test circuit for tables XI and XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PWA = 200 ±25 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. All voltages are referenced to ground.

FIGURE 16 Static output voltage (low-level and high-level, V_{DD} = 15 V) test circuit waveform for tables XI and XII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XIII. Input current (low-level) test.

Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L1	1R	Input current, low level	25,19,40	6	0	-200	nA	17
L2	1Ø	↓	6,19,40	25	↓	↓	↓	↓
L3	2R		6,25,40	19				
L4	2Ø		6,25,19	40				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

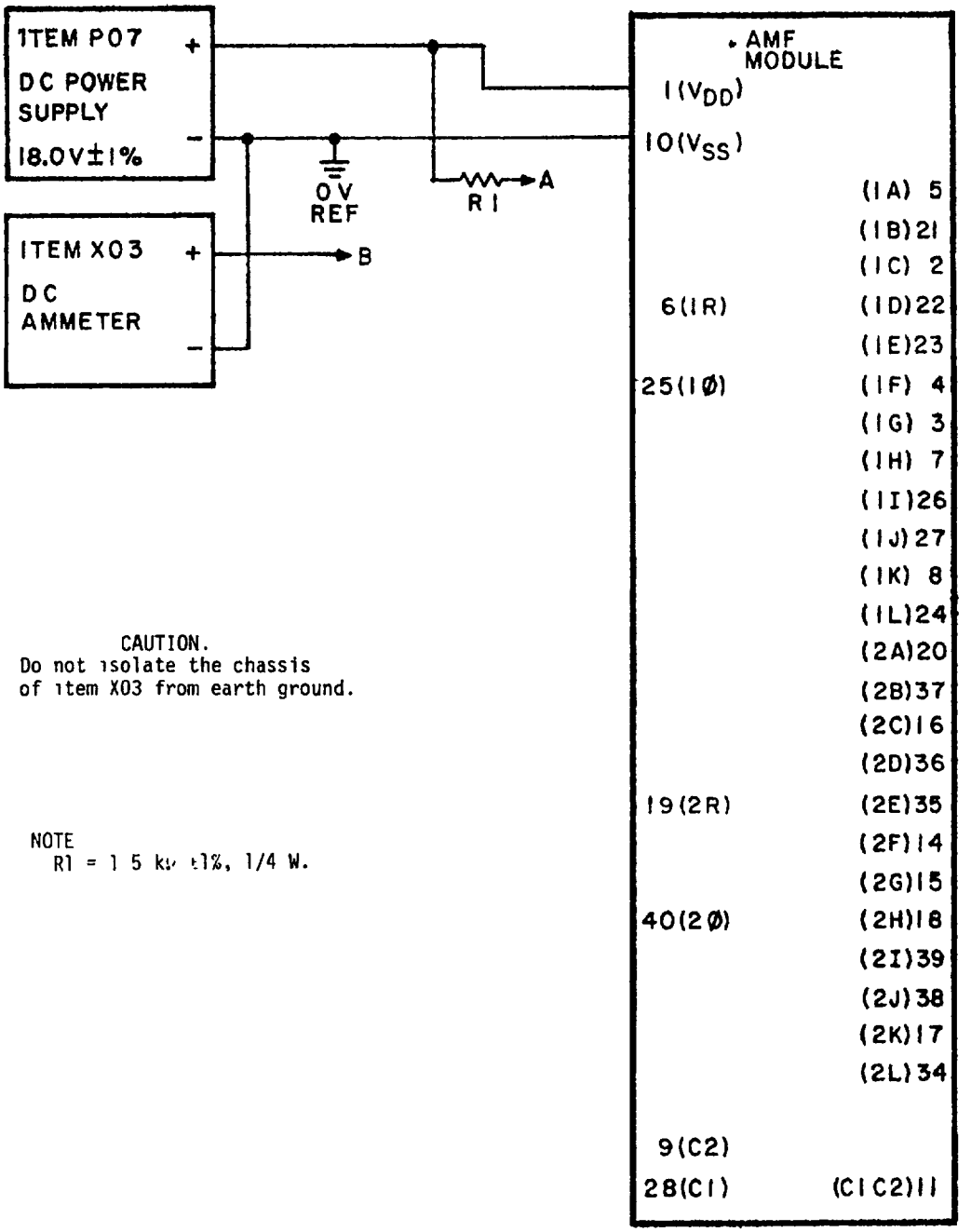


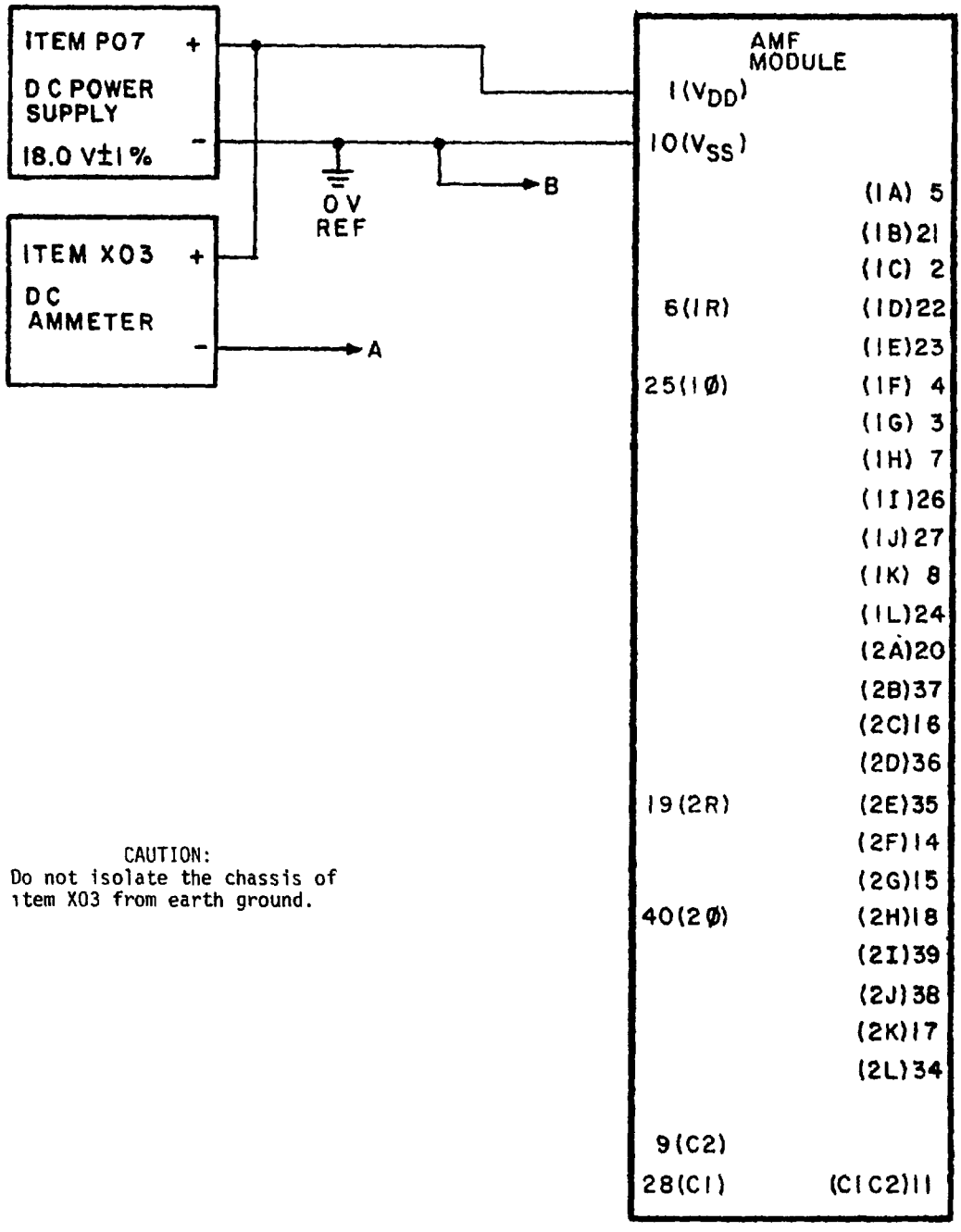
FIGURE 17. Input current (low-level) test circuit for table XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XIV. Input current (high-level) test.

Test condition	Parameter		Connect: A to pin	Connect: B to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
M1	1R	Input current, high level	6	25,19,40	0	200	nA	18
M2	1Ø	↓	25	6,19,40	↓	↓	↓	↓
M3	2R		19	6,25,40				
M4	2Ø		40	6,25,19				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



CAUTION:
Do not isolate the chassis of item X03 from earth ground.

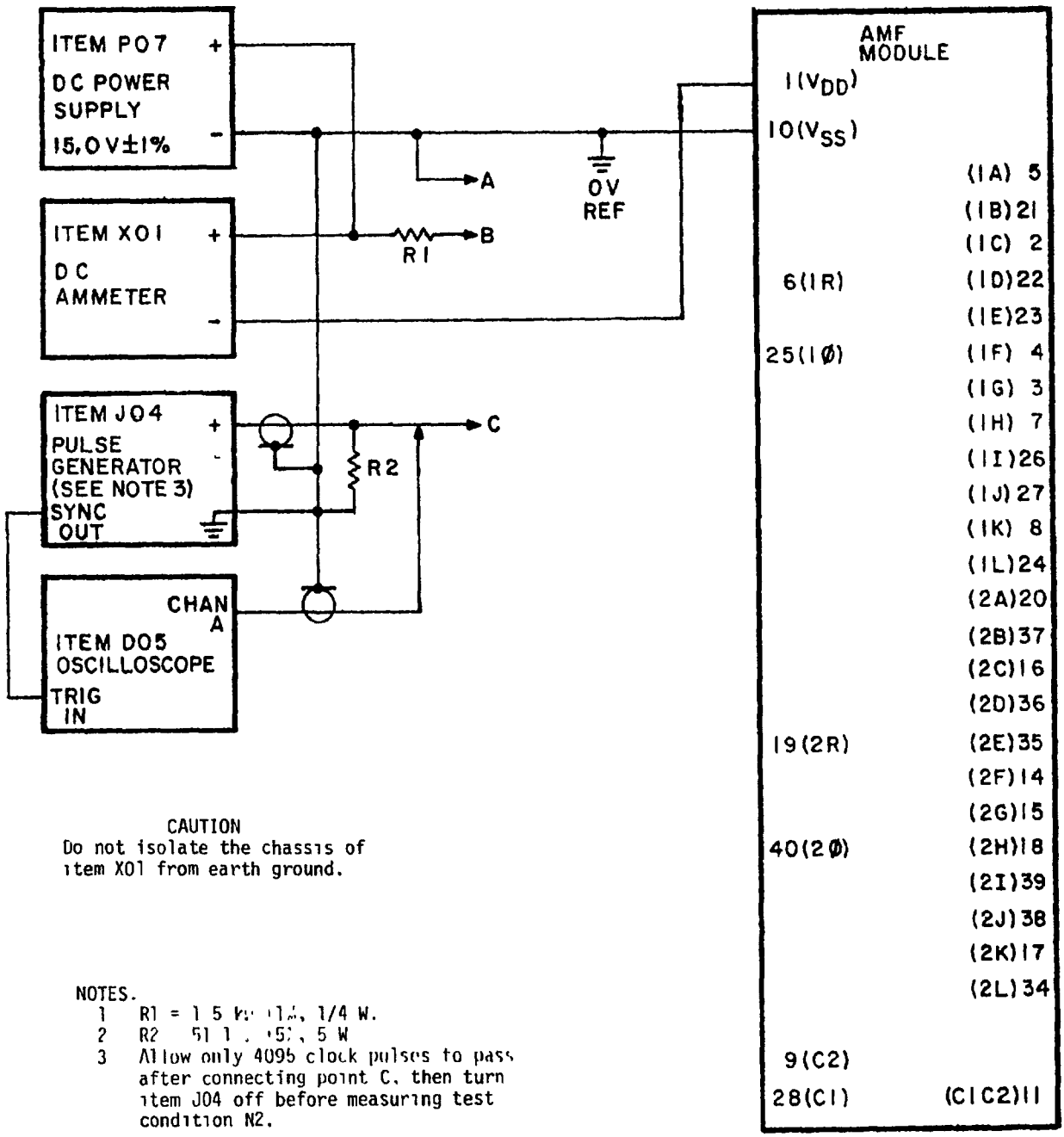
FIGURE 18. Input current (high-level) test circuit for table XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XV. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	Power supply current, low level	Open	6,25,19,40	Open	0	40	µA	19,20
M2	Power supply current, high level	6,19	Open	25,40	0	40	µA	19,20

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

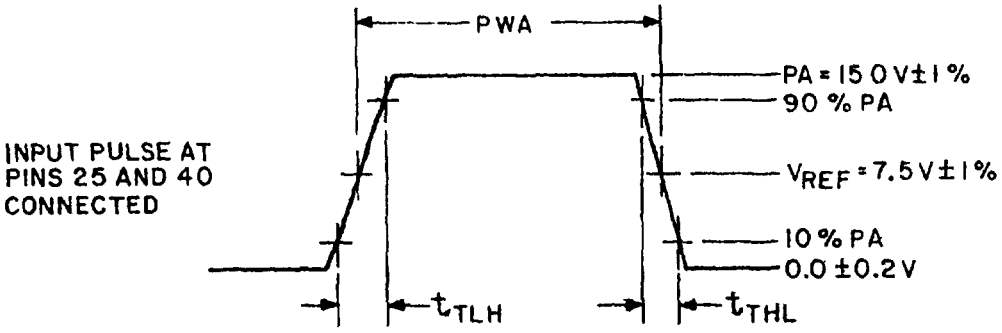


CAUTION
Do not isolate the chassis of item X01 from earth ground.

NOTES.
1. R1 = 1.5 kΩ (1%, 1/4 W.
2. R2 = 51 Ω (5%, 5 W
3. Allow only 4095 clock pulses to pass after connecting point C, then turn item J04 off before measuring test condition N2.

FIGURE 19. Power supply current (low-level and high-level) test circuit for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES.
1. Input pulse parameters
   PWA = 500 +50 ns
   $t_{THL} - t_{TLH}$ = 10 2 ns
2. All voltages are referenced to ground

FIGURE 20. Power supply current (low-level and high-level) test circuit waveform for table XV

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XVI. Transition time (t_{TLH}) test.

Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
P1	1A	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	Open	5	35	350	ns	21,22
P2	1B			21				
P3	1C			2				
P4	1D			22				
P5	2A			20				
P6	2B			37				
P7	2C			16				
P8	2D			36				
P9	1A	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V		5	83	830		
P10	1B			21				
P11	1C			2				
P12	1D			22				
P13	2A			20				
P14	2B			37				
P15	2C			16				
P16	2D			36				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XVII. Transition time (t_{THL}) test.

Test condition		Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
Q1	1A	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	Open	5	17	175	ns	21,22
Q2	1B			21				
Q3	1C			2				
Q4	1D			22				
Q5	2A			20				
Q6	2B			37				
Q7	2C			16				
Q8	2D			36				
Q9	1A	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V		5	65	655		
Q10	1B			21				
Q11	1C			2				
Q12	1D			22				
Q13	2A			20				
Q14	2B			37				
Q15	2C			16				
Q16	2D			36				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Transition time (t_{TLH}) test.

Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
R1	1A	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	Open	5	15	150	ns	21,22
R2	1B	↓	↓	21	↓	↓	↓	↓
R3	1C			2				
R4	1D			22				
R5	2A			20				
R6	2B			37				
R7	2C			16				
R8	2D			36				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XIX. Transition time (t_{THL}) test.

Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
S1	1A	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	Open	5	7	75	ns	21,22
S2	1B			21				
S3	1C			2				
S4	1D			22				
S5	2A			20				
S6	2B			37				
S7	2C			16				
S8	2D			36				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XX. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T1	1A Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	Open	5	11	110	ns	21, 22
T2	1B		21				
T3	1C		2				
T4	1D		22				
T5	2A		20				
T6	2B		37				
T7	2C		16				
T8	2D		36				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XXI. Transition time (t_{THL}) test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
U1	1A	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	Open	5	5	55	ns	21,22
U2	1B			21				
U3	1C			2				
U4	1D			22				
U5	2A			20				
U6	2B			37				
U7	2C			16				
U8	2D			36				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XXII. Propagation delay time (t_{PLH}) test.

Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
V1	1A	Propagation delay time (t_{PLH}) C_L = 50 pF, V_{DD} = 5 V	25	5	70	700	ns	21,22
V2	1B		5	21	40	391		
V3	1C		21	2	40	391		
V4	1D		2	22	40	391		
V5	2A		40	20	70	700		
V6	2B		20	37	40	391		
V7	2C		37	16	40	391		
V8	2D		16	36	40	391		
V9	1A	Propagation delay time (t_{PLH}) C_L = 150 pF, V_{DD} = 5 V	25	5	90	900		
V10	1B		5	21	60	591		
V11	1C		21	2	60	591		
V12	1D		2	22	60	591		
V13	2A		40	20	90	900		
V14	2B		20	37	60	591		
V15	2C		37	16	60	591		
V16	2D		16	36	60	591		

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XXIII. Propagation delay time (t_{PHL}) test.

Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
W1	1A	Propagation delay time (t_{PLH}) C_L = 50 pF, V_{DD} = 5 V	25	5	70	700	ns	21,22
W2	1B		5	21	40	391		
W3	1C		21	2	40	391		
W4	1D		2	22	40	391		
W5	2A		40	20	70	700		
W6	2B		20	37	40	391		
W7	2C		37	16	40	391		
W8	2D		16	36	40	391		
W9	1A	Propagation delay time (t_{PLH}) C_L = 150 pF, V_{DD} = 5 V	25	5	90	900		
W10	1B		5	21	60	591		
W11	1C		21	2	60	591		
W12	1D		2	22	60	591		
W13	2A		40	20	90	900		
W14	2B		20	37	60	591		
W15	2C		37	16	60	591		
W16	2D		16	36	60	591		

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XXIV. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X1	1A Propagation delay time (t_{PLH}) C_L = 50 pF, V_{DD} = 10 V	25	5	28	280	ns	21,22
X2	1B	5	21	14	138		
X3	1C	21	2	14	138		
X4	1D	2	22	14	138		
X5	2A	40	20	28	280		
X6	2B	20	37	14	138		
X7	2C	37	16	14	138		
X8	2D	16	36	14	138		

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XXV. Propagation delay time (t_{PHL}) test.

Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
Y1	1A	Propagation delay time (t_{PHL}) C_L = 50 pF, V_{DD} = 10 V	25	5	28	280	ns	21,22
Y2	1B		5	21	14	138		
Y3	1C		21	2	14	138		
Y4	1D		2	22	14	138		
Y5	2A		40	20	28	280		
Y6	2B		20	37	14	138		
Y7	2C		37	16	14	138		
Y8	2D		16	36	14	138		

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XXVI. Propagation delay time (t_{PLH}) test.

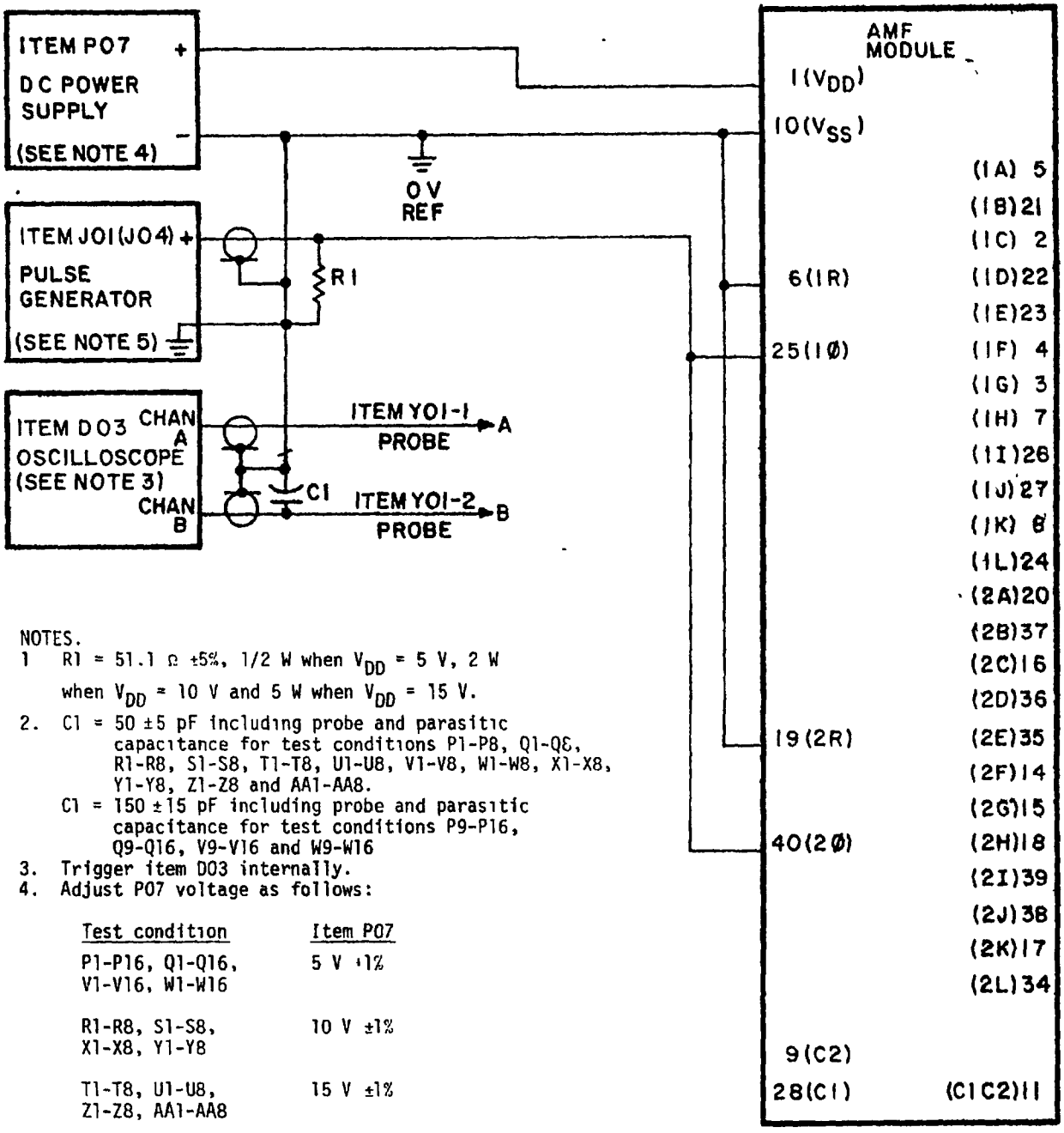
Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
Z1	1A	Propagation delay time (t_{PLH}) C_L = 50 pF, V_{DD} = 15 V	25	5	21	210	ns	21, 22
Z2	1B	↓	5	21	11	108	↓	↓
Z3	1C		21	2	11	108		
Z4	1D		2	22	11	108		
Z5	2A		40	20	21	210		
Z6	2B		20	37	11	108		
Z7	2C		37	16	11	108		
Z8	2D		16	36	11	108		

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XXVII. Propagation delay time (t_{PHL}) test.

Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
AA1	1A	Propagation delay time (t_{PHL}) C_L = 50 pF, V_{DD} = 15 V	25	5	21	210	ns	21,22
AA2	1B		5	21	11	108		
AA3	1C		21	2	11	108		
AA4	1D		2	22	11	108		
AA5	2A		40	20	21	210		
AA6	2B		20	37	11	108		
AA7	2C		37	16	11	108		
AA8	2D		16	36	11	108		

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES.

1. R1 = 51.1 Ω ±5%, 1/2 W when V_{DD} = 5 V, 2 W when V_{DD} = 10 V and 5 W when V_{DD} = 15 V.
2. C1 = 50 ±5 pF including probe and parasitic capacitance for test conditions P1-P8, Q1-Q8, R1-R8, S1-S8, T1-T8, U1-U8, V1-V8, W1-W8, X1-X8, Y1-Y8, Z1-Z8 and AA1-AA8.
   C1 = 150 ±15 pF including probe and parasitic capacitance for test conditions P9-P16, Q9-Q16, V9-V16 and W9-W16
3. Trigger item D03 internally.
4. Adjust P07 voltage as follows:

Test condition	Item P07
P1-P16, Q1-Q16, V1-V16, W1-W16	5 V ±1%
R1-R8, S1-S8, X1-X8, Y1-Y8	10 V ±1%
T1-T8, U1-U8, Z1-Z8, AA1-AA8	15 V ±1%

5. Use item J01 for test conditions P1-P16, Q1-Q16, R1-R16, S1-S8, V1-V16, W1-W16, X1-X8 and Y1-Y8 and item J04 for test conditions T1-T8, U1-U8, Z1-Z8 and AA1-AA8.

FIGURE 21. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XVI through XXVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

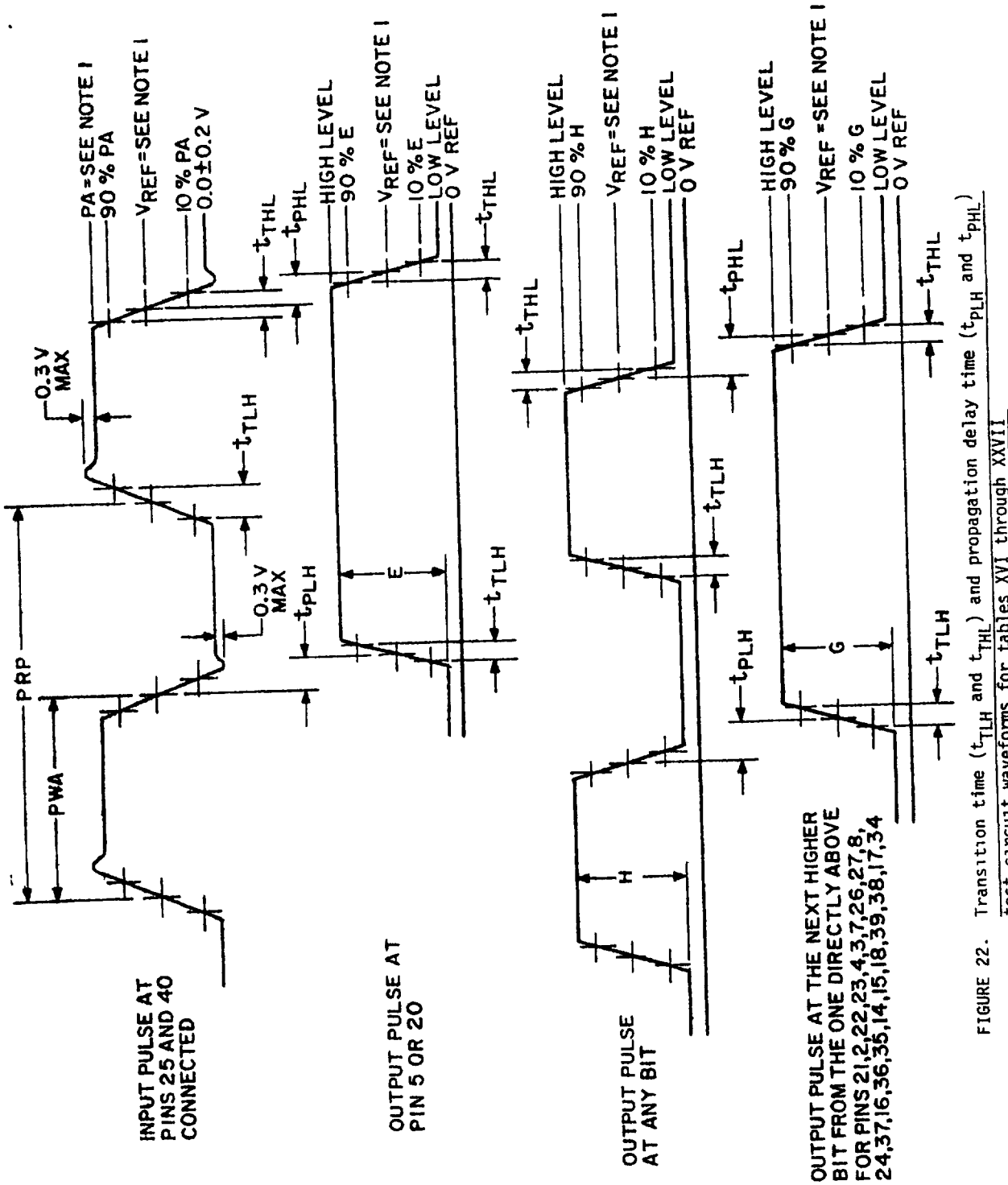


FIGURE 22. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVI through XXVII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

NOTES:
1. Input pulse parameters:
   PWA = 1.0 ±0.1 µs.
   PRP = 2.0 ±0.2 µs.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns
   PA = 5.0 ±0.1 V for test conditions P1-P16, Q1-Q16, V1-V16, W1-W16.
   = 10.0 ±0.1 V for test conditions R1-R8, S1-S8, X1-X8, Y1-Y8
   = 15.0 ±0.1 V for test conditions T1-T8, U1-U8, Z1-Z8, AA1-AA8.
   V_{REF} = 2.5 ±0.1 V for test conditions P1-P16, Q1-Q16, V1-V16, W1-W16.
   = 5.0 ±0.1 V for test conditions R1-R8, S1-S8, X1-X8, Y1-Y8.
   = 7.5 ±0.1 V for test conditions T1-T8, U1-U8, Z1-Z8, AA1-AA8
2. Before making the measurements, verify that the output pulse high level and low level are as follows

V_{DD}	High level	Low level
5 V	≥ 4.6 V	≤ 0.4 V
10 V	≥ 9.5 V	≤ 0.5 V
15 V	≥13.5 V	≤ 1.5 V

3. The lead length of electrical conductors between terminator resistor R1 and output pin, probes Y01-1 and Y01-2 to input and output pins and between load circuit and output pins shall be less than one inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 22. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVI through XXVII- Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XXVIII. Propagation delay time (t_{PHL}) test.

Test condition		Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
BB1	1A	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	5	108	1080	ns	23,24
BB2	1B		21				
BB3	1C		2				
BB4	1D		22				
BB5	1E		23				
BB6	1F		4				
BB7	1G		3				
BB8	1H		7				
BB9	1I		26				
BB10	1J		27				
BB11	1K		8				
BB12	1L		24				
BB13	2A		20				
BB14	2B		37				
BB15	2C		16				
BB16	2D		36				
BB17	2E		35				
BB18	2F		14				
BB19	2G		15				
BB20	2H		18				
BB21	2I		39				
BB22	2J		38				
BB23	2K		17				
BB24	2L		34				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XXVIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition		Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
BB25	1A	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	5	128	1280	ns	23,24
BB26	1B		21				
BB27	1C		2				
BB28	1D		22				
BB29	1E		23				
BB30	1F		4				
BB31	1G		3				
BB32	1H		7				
BB33	1I		26				
BB34	1J		27				
BB35	1K		8				
BB36	1L		24				
BB37	2A		20				
BB38	2B		37				
BB39	2C		16				
BB40	2D		36				
BB41	2E		35				
BB42	2F		14				
BB43	2G		15				
BB44	2H		18				
BB45	2I		39				
BB46	2J		38				
BB47	2K		17				
BB48	2L		34				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XXIX. Propagation delay time (t_{PHL}) test.

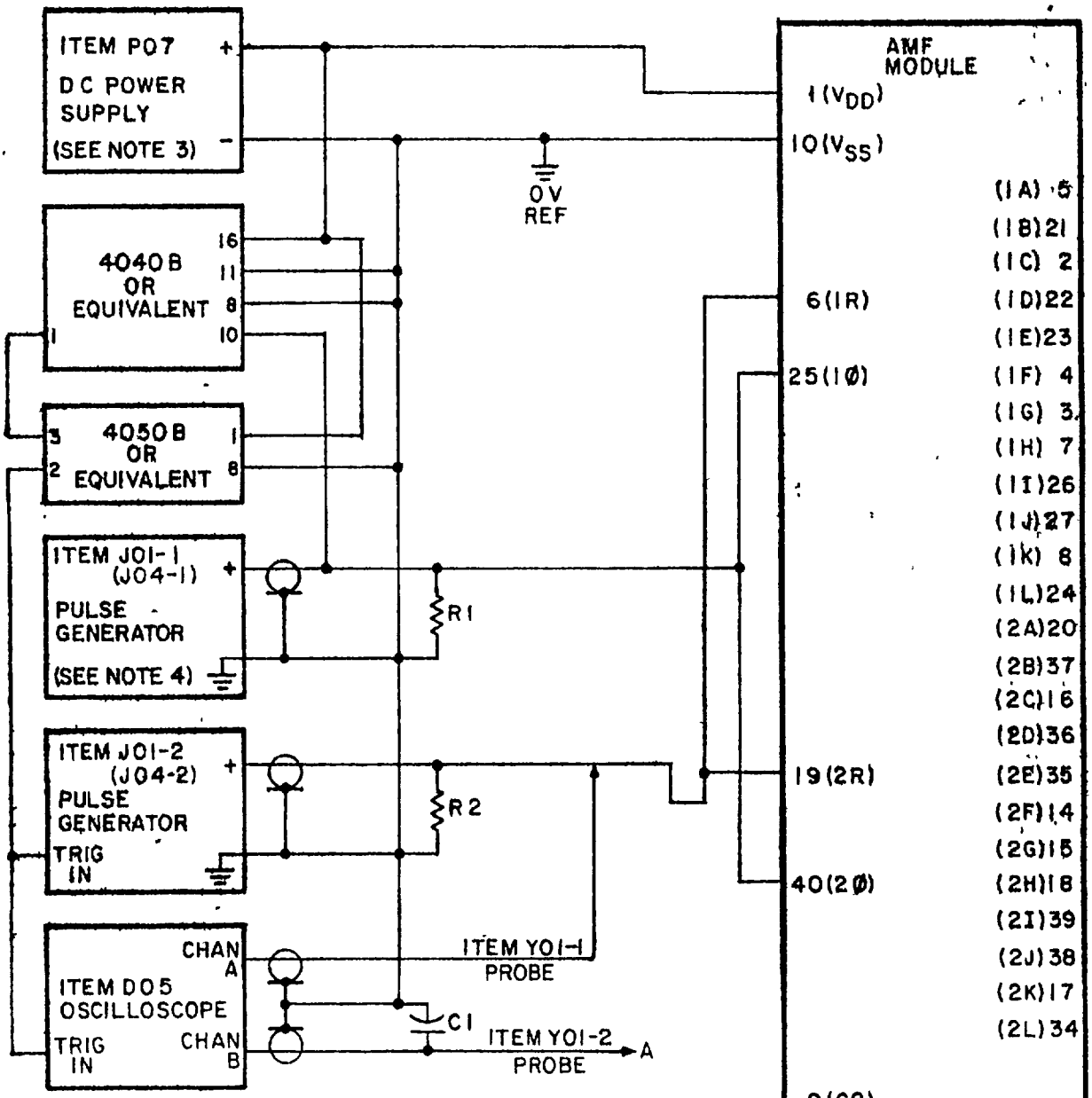
Test condition		Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
CC1	1A	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	5	40	400	ns	23,24
CC2	1B		21				
CC3	1C		2				
CC4	1D		22				
CC5	1E		23				
CC6	1F		4				
CC7	1G		3				
CC8	1H		7				
CC9	1I		26				
CC10	1J		27				
CC11	1K		8				
CC12	1L		24				
CC13	2A		20				
CC14	2B		37				
CC15	2C		16				
CC16	2D		36				
CC17	2E		35				
CC18	2F		14				
CC19	2G		15				
CC20	2H		18				
CC21	2I		39				
CC22	2J		38				
CC23	2K		17				
CC24	2L		34				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XXX. Propagation delay time (t_{PHL}) test.

Test condition		Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
DD1	1A	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	5	30	300	ns	3,24
DD2	1B		21				
DD3	1C		2				
DD4	1D		22				
DD5	1E		23				
DD6	1F		4				
DD7	1G		3				
DD8	1H		7				
DD9	1I		26				
DD10	1J		27				
DD11	1K		8				
DD12	1L		24				
DD13	2A		20				
DD14	2B		37				
DD15	2C		16				
DD16	2D		36				
DD17	2E		35				
DD18	2F		14				
DD19	2G		15				
DD20	2H		18				
DD21	2I		39				
DD22	2J		38				
DD23	2K		17				
DD24	2L		34				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES.

1. R1 = R2 = 51.1 Ω ±5%, 1/2 W when V_{DD} = 5 V, 2 W when V_{DD} = 10 V and 5 W when V_{DD} = 15 V.
2. C1 = 50 ±5 pF including probe and parasitic capacitance for test conditions BB1-BB24, CC1-CC24 and DD1-DD24.
   C1 = 150 ±15 pF including probe and parasitic capacitance for test conditions BB25-BB48.
3. Adjust item P07 voltage as follows:

Test conditions	Item P07
BB1-BB48	5 V ±1%
CC1-CC24	10 V ±1%
DD1-DD24	15 V ±1%

4. Use item J01 for test conditions BB1-BB48 and CC1-CC24 and item J04 for test conditions DD1-DD24.

FIGURE 23. Propagation delay time (t_{PHL}) test circuit for tables XXVIII, XXIX and XXX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

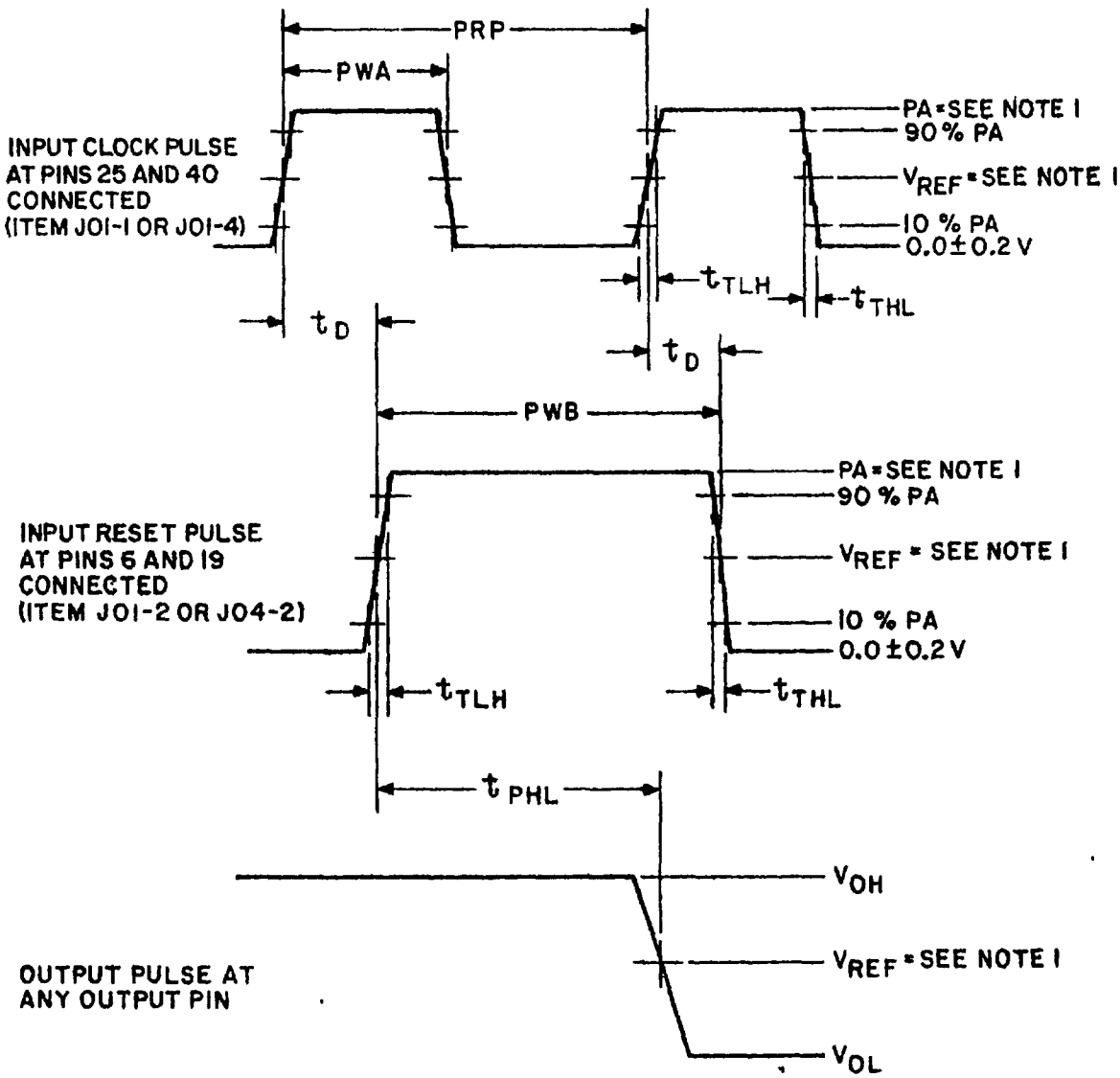


FIGURE 24. Propagation delay time (t_{PHL}) test circuit waveforms for tables XXVIII, XXIX and XXX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

NOTES:

1. Input pulse parameters:
   PWA = 1000 ±100 ns
   PWB = 2000 ±200 ns
   PRP = 2000 ±200 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PA = 5.0 ±0.1 V for test conditions BB1-BB48
   = 10.0 ±0.1 V for test conditions CC1-CC24
   = 15 0 ±0 1 V for test conditions DD1-DD24.
   V_{REF} = 2 5 ±0.1 V for test conditions BB1-BB48
   = 5.0 ±0.1 V for test conditions CC1-CC24.
   = 7.5 ±0.1 V for test conditions DD1-DD24.
2. t_D = 500 ±50 ns.
3. Before making the measurements, verify that the output pulse high level and low level are as follows:

V_{DD}	High level	Low level
5 V	≥ 4.6 V	≤ 0 4 V
10 V	≥ 9.5 V	≤ 0.5 V
15 V	≥13 5 V	≤ 1.5 V

4. The lead length of electrical conductors between terminator resistors R1 and R2 and input pins, probes Y01-1 and Y01-2 and input pins shall be less than one inch (25.4 mm)
5. All voltages are referenced to ground

FIGURE 24. Propagation delay time (t_{PHL}) test circuit waveforms for tables XXVIII, XXIX and XXX - Continued

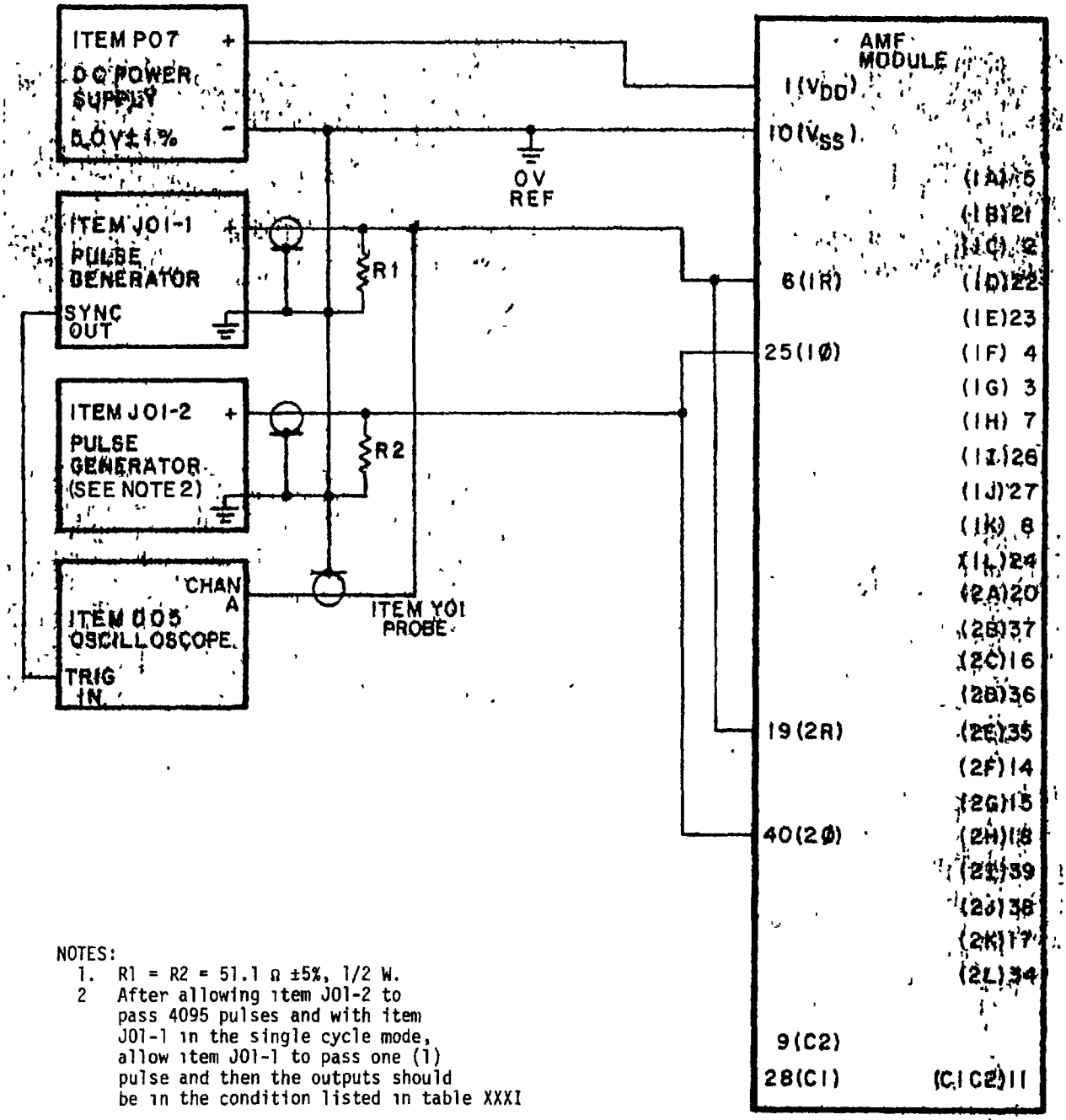
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XXXI. Minimum reset pulse width check test.

Test condition	Parameter	Output levels 1/																								Test figure
		1A	1B	1C	1D	1E	1F	1G	1H	1I	1J	1K	1L	2A	2B	2C	2D	2E	2F	2G	2H	2I	2J	2K	2L	
EE	Minimum reset pulse width check	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	25,26

1/ L = low level = 0.0 V min, 0.4 V max.

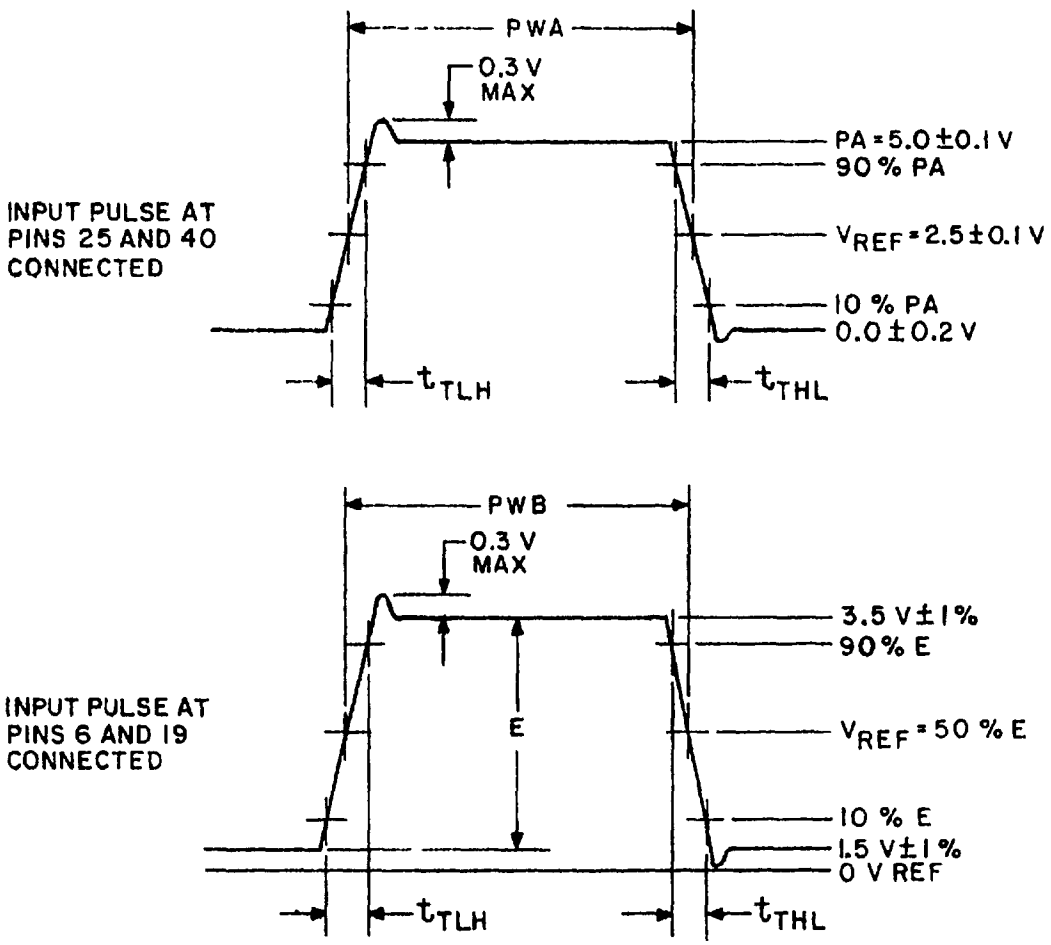
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51.1 Ω ±5%, 1/2 W.
2 After allowing item J01-2 to pass 4095 pulses and with item J01-1 in the single cycle mode, allow item J01-1 to pass one (1) pulse and then the outputs should be in the condition listed in table XXXI

FIGURE 25. Minimum reset pulse width check test circuit for table XXXI

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES:

1. Input pulse parameters for item J01-2:
   PWA = 1000 ±100 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. Input pulse parameters for item J01-1.
   PWB = 640^{+60}_{-0} ns.
3. All voltages are referenced to ground

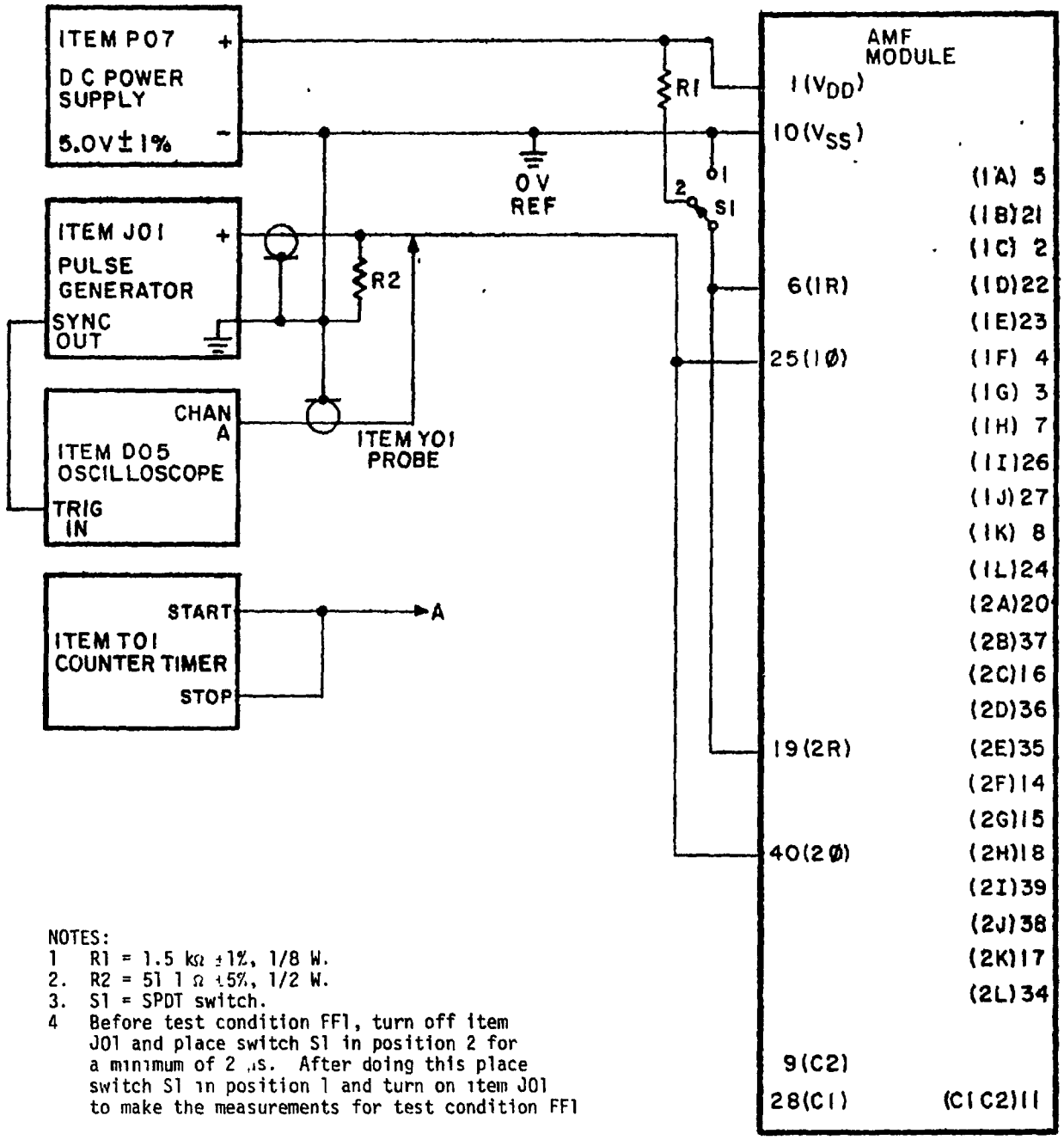
FIGURE 26. Minimum reset pulse width check test circuit waveforms for table XXXI

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XXXII. Maximum clock frequency check test.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
FF1	1L	Maximum clock frequency check	24	1.95	2.15	ms	27,28
FF2	2L	Maximum clock frequency check	34	1.95	2.15	ms	27,28

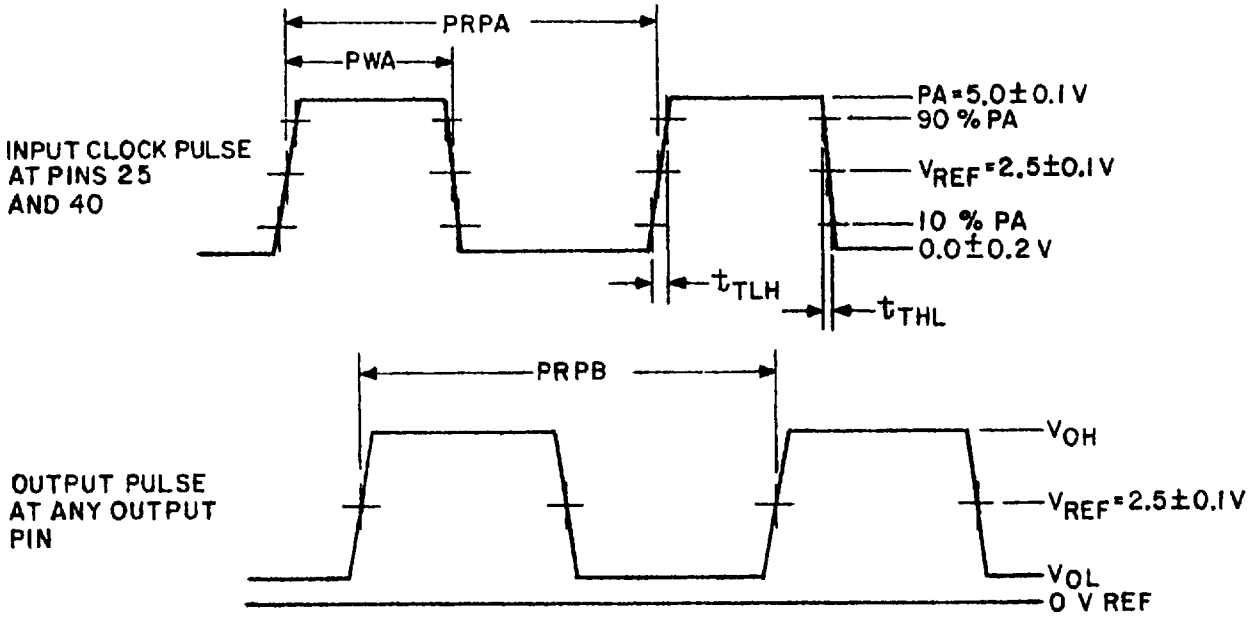
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 1.5 kΩ ±1%, 1/8 W.
2. R2 = 51 1 Ω ±5%, 1/2 W.
3. S1 = SPDT switch.
4. Before test condition FF1, turn off item J01 and place switch S1 in position 2 for a minimum of 2 µs. After doing this place switch S1 in position 1 and turn on item J01 to make the measurements for test condition FF1

FIGURE 27 Maximum clock frequency check test circuit for table XXXII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters

   PWA = 255 ±5 ns.
   PRPA = 510 ±10 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.

2. All voltages are referenced to ground
3. Before making any measurements, verify that the output pulse high level is $\geq$ 4.6 V and the low level is $\leq$ 0.4 V.
4. The lead length of electrical conductors between terminator resistor R2 and input pin, probe YO1 to input pin shall be less than one inch (25.4 mm).

FIGURE 28. Maximum clock frequency check test circuit waveforms for table XXXII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XXXIII. Power supply filter capacitance test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
GG	Power supply filter capacitance	Open	Open	1	0.007	0.013	µF	20

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XXXIV. Capacitance (C1 and C2) test.

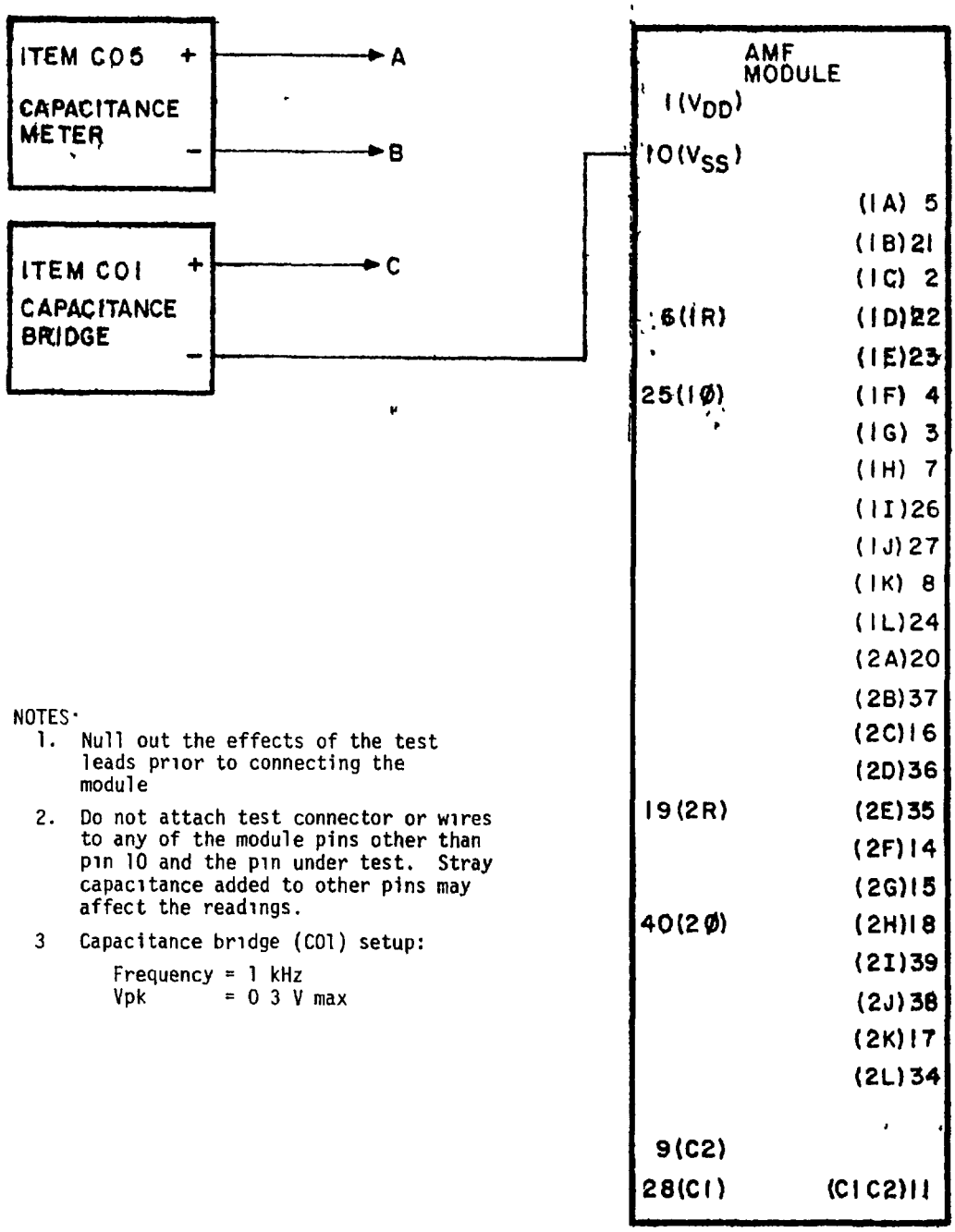
Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
HH1	C1	Capacitance	28	11	Open	176	264	pF	29
HH2	C2	Capacitance	9	11	Open	176	264	pF	29

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

TABLE XXXV. Input capacitance test.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
JJ1	1R	Input capacitance	6	10	Open	2	15	pF	29
JJ2	1Ø	↓	25	↓	↓	↓	↓	↓	↓
JJ3	2R		19						
JJ4	2Ø		40						

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.



NOTES:

1. Null out the effects of the test leads prior to connecting the module
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup:

   Frequency = 1 kHz
   Vpk = 0 3 V max

FIGURE 29 Power supply filter capacitance, capacitance (C1 and C2) and input capacitance test circuit for tables XXXIII, XXXIV and XXXV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE: MIL-M-28787/250(EC), COUNTER, BINARY, DIGITAL, KEY CODE AMF

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Electronic Systems Command
Attn; ELEX 8111
Washington, DC 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:03Z
Check the source to verify that this is the current version before use.